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Basic DESeq2 results exploration

Comparison: Control vs Treated

Introduction

This report is meant to help explore DESeq2 (Love, Huber, and Anders, 2014) results and was generated using the `regionReport` (Collado-Torres, Jaffe, and Leek, 2016) package. While the report is rich, it is meant to just start the exploration of the results and exemplify some of the code used to do so. If you need a more in-depth analysis for your specific data set you might want to use the `customCode` argument. This report is based on the vignette of the `DESeq2` (Love, Huber, and Anders, 2014) package which you can find here.

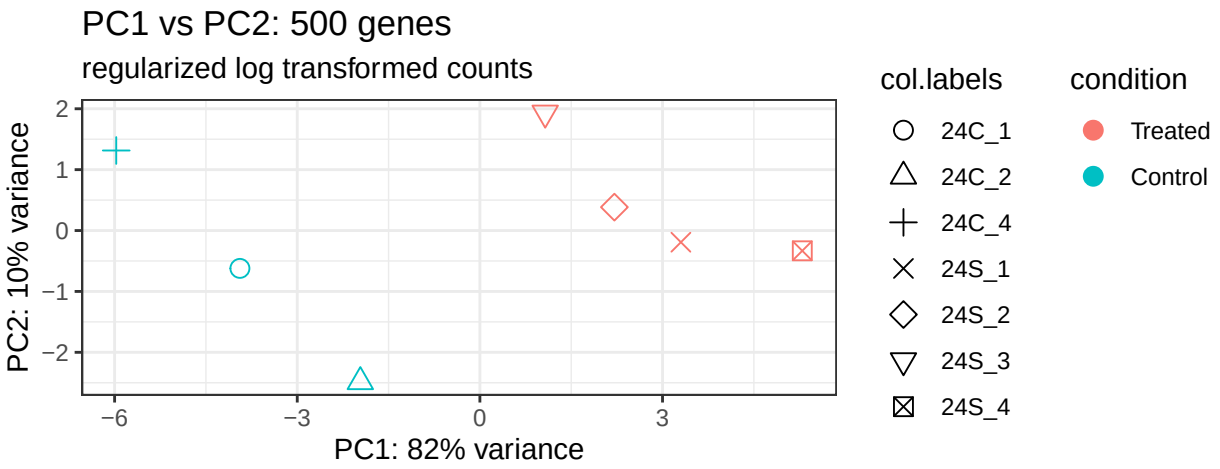
Code setup

This section contains the code for setting up the rest of the report.

```
## knitrBoostrap and device chunk options
library('knitr')
opts_chunk$set(bootstrap.show.code = FALSE, dev = device, crop = NULL)
if(!outputIsHTML) opts_chunk$set(bootstrap.show.code = FALSE, dev = device, echo = FALSE)
```

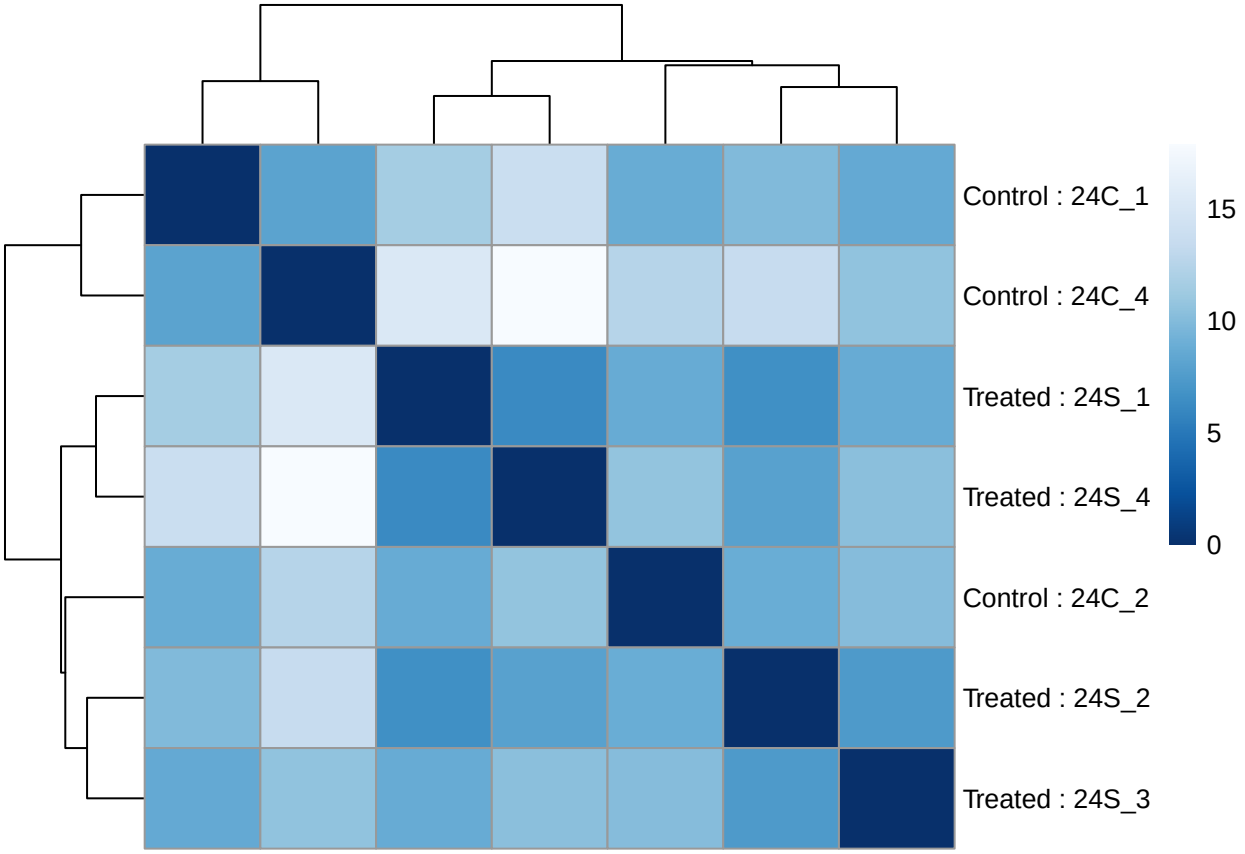
PCA

The below plot shows the first two principal components that explain the variability in the data using the regularized log (`rlog`) count data. If you are unfamiliar with principal component analysis, you might want to check the Wikipedia entry or this interactive explanation.



In this case (`rlog` count data), the first and second principal component explain 82 and 10 percent of the variance respectively.

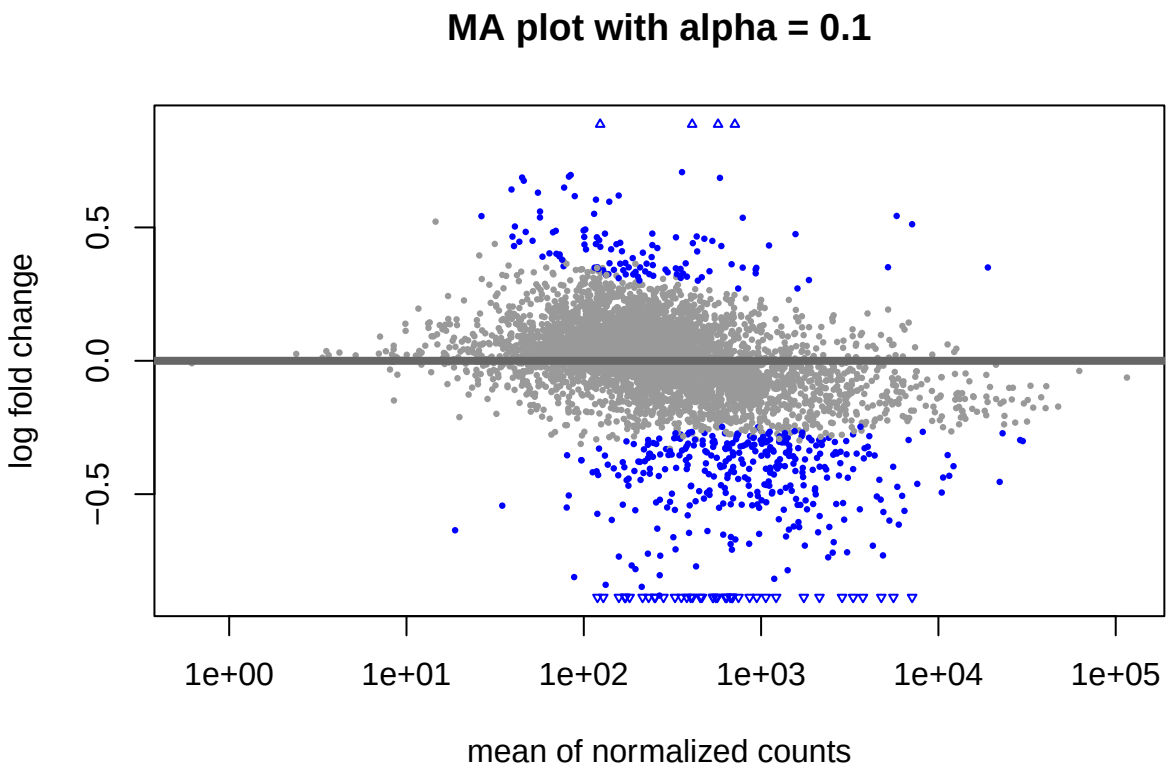
Sample-to-sample distances



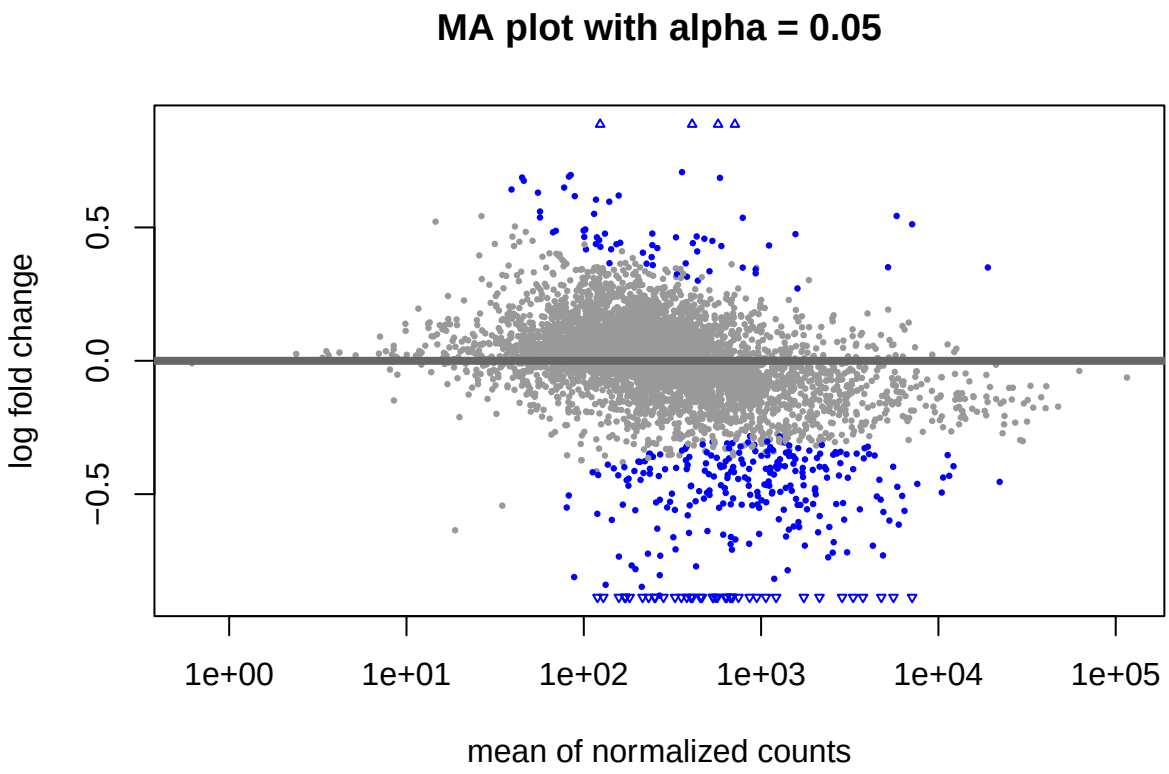
This plot shows how samples are clustered based on their euclidean distance using the regularized log transformed count data. This figure gives an overview of how the samples are hierarchically clustered. It is a complementary figure to the PCA plot.

MA plots

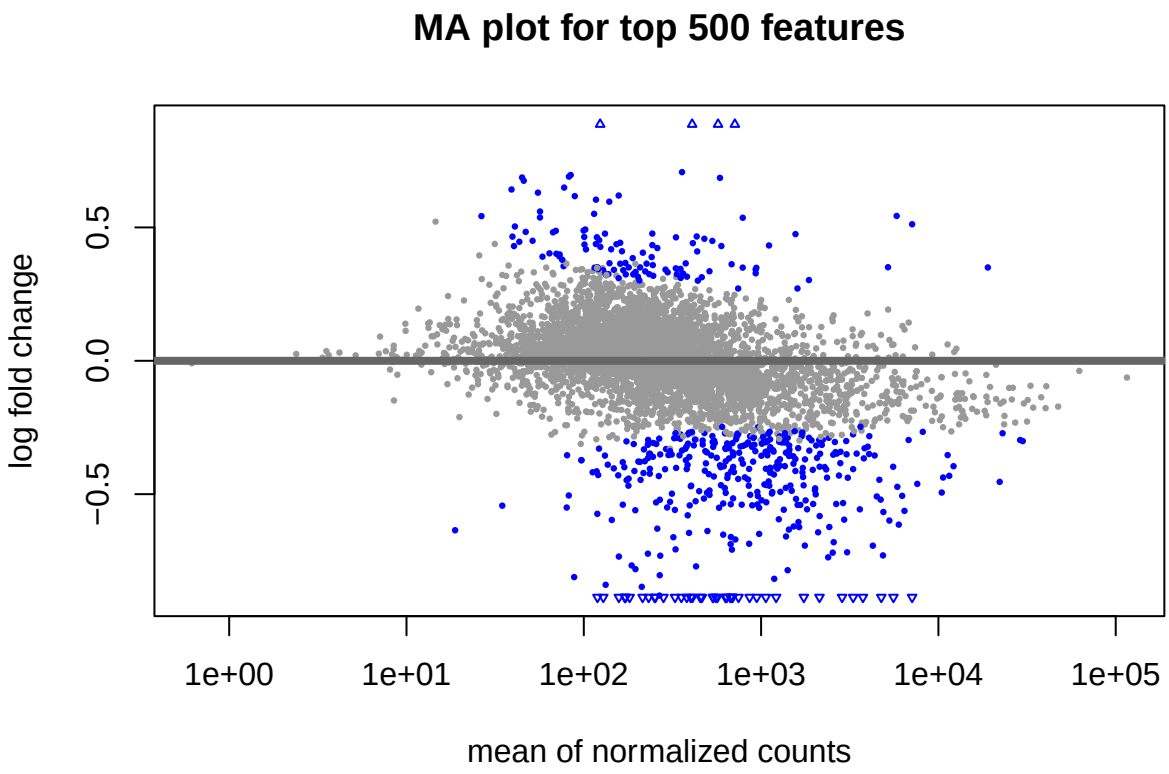
This section contains three MA plots (see Wikipedia) that compare the mean of the normalized counts against the log fold change. They show one point per feature. The points are shown in red if the feature has an adjusted p-value less than `alpha`, that is, the statistically significant features are shown in red.



This first plot shows uses `alpha` = 0.1, which is the `alpha` value used to determine which resulting features were significant when running the function `DESeq2::results()`.



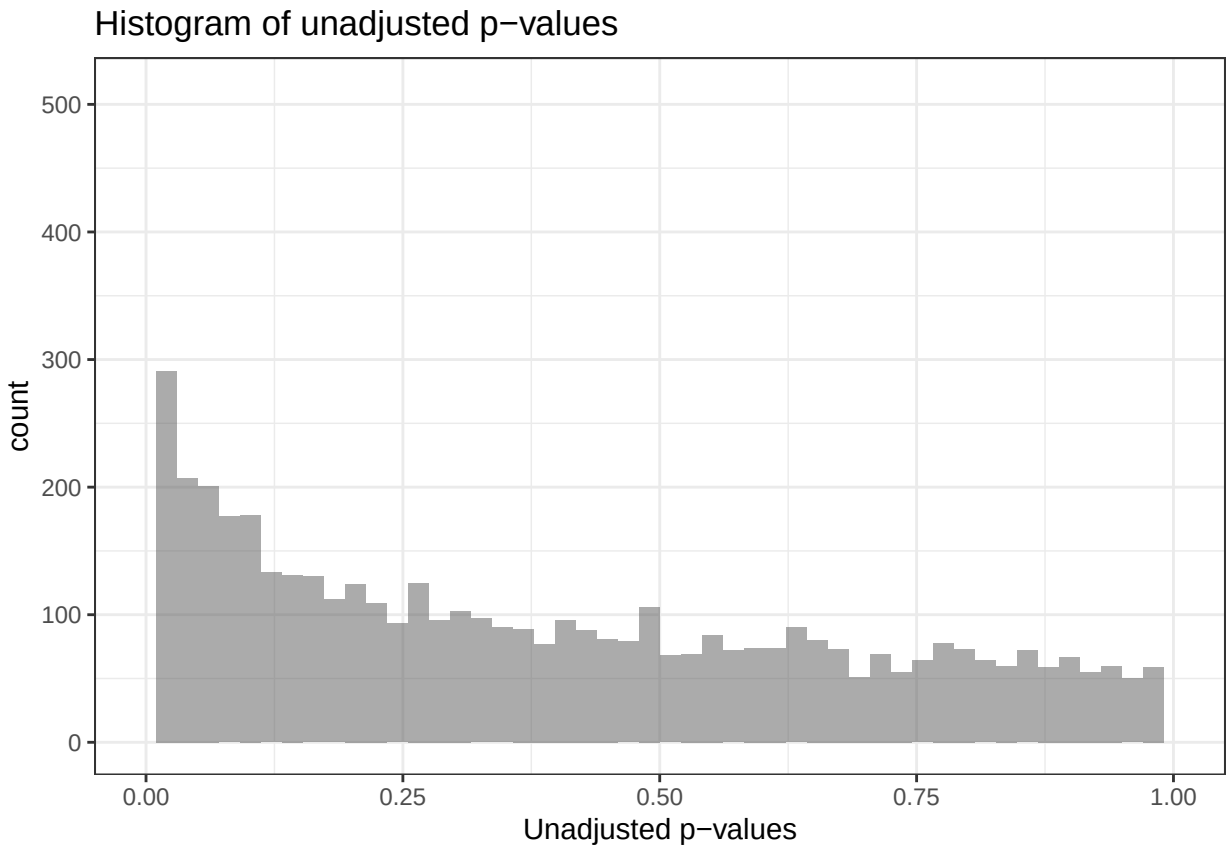
This second MA plot uses `alpha` = 0.05 and can be used agains the first MA plot to identify which features have adjusted p-values between 0.05 and 0.1.



The third and final MA plot uses an alpha such that the top 500 features are shown in the plot. These are the features that whose details are included in the *top features* interactive table.

P-values distribution

```
## Warning: Removed 2 rows containing missing values (geom_bar).
```



This plot shows a histogram of the unadjusted p-values. It might be skewed right or left, or flat as shown in the Wikipedia examples. The shape depends on the percent of features that are differentially expressed. For further information on how to interpret a histogram of p-values check David Robinson's post on this topic.

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.08133 0.28975 0.35691 0.59765 0.99939
```

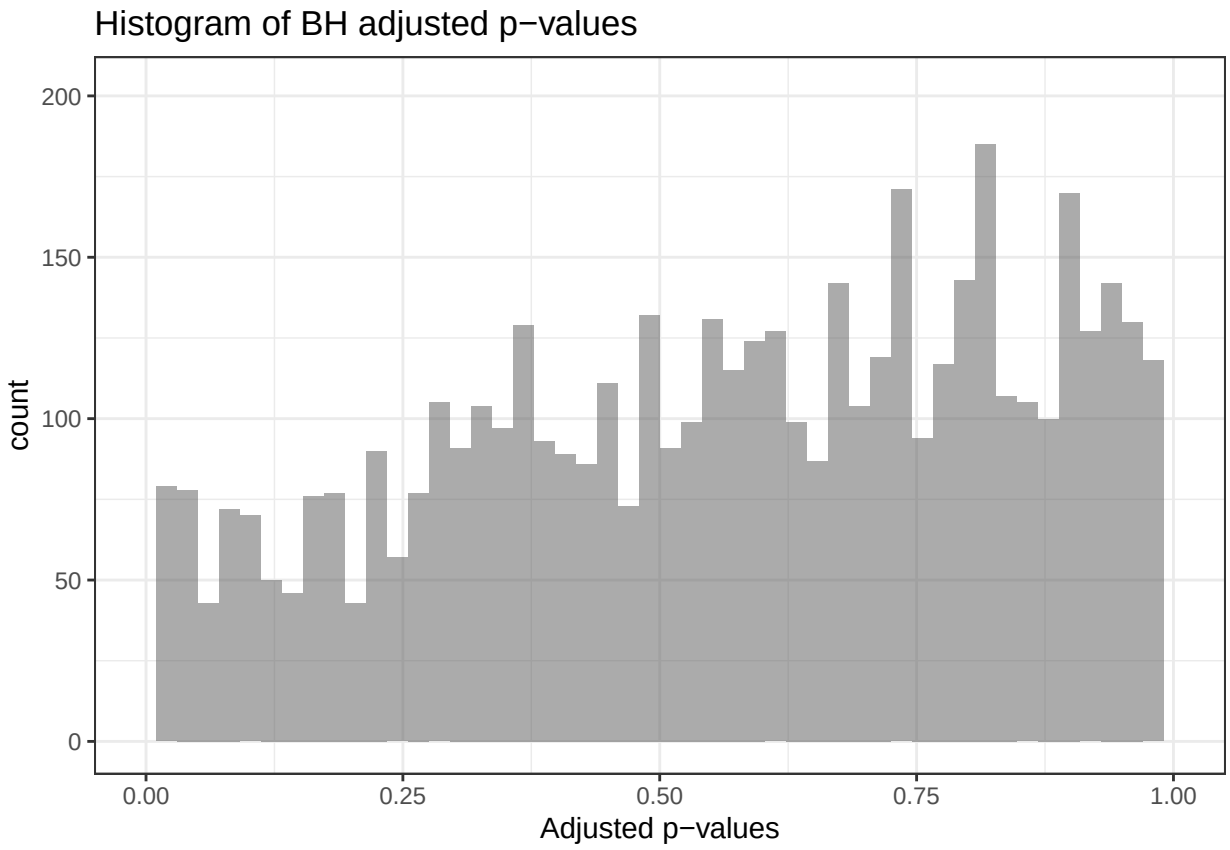
This is the numerical summary of the distribution of the p-values.

Cut	Count
0.0001	160
0.0010	244
0.0100	507
0.0250	729
0.0500	1003
0.1000	1455
0.2000	2114
0.3000	2647
0.4000	3084
0.5000	3522
0.6000	3886
0.7000	4247
0.8000	4576
0.9000	4888
1.0000	5174

This table shows the number of features with p-values less or equal than some commonly used cutoff values.

Adjusted p-values distribution

```
## Warning: Removed 2 rows containing missing values (geom_bar).
```



This plot shows a histogram of the BH adjusted p-values. It might be skewed right or left, or flat as shown in the Wikipedia examples.

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.3245  0.5792  0.5475  0.7968  0.9994
```

This is the numerical summary of the distribution of the BH adjusted p-values.

Cut	Count
0.0001	84
0.0010	124
0.0100	201
0.0250	259
0.0500	353
0.1000	501
0.2000	810
0.3000	1180
0.4000	1685
0.5000	2170
0.6000	2716
0.7000	3270

Cut	Count
0.8000	3890
0.9000	4554
1.0000	5174

This table shows the number of features with BH adjusted p-values less or equal than some commonly used cutoff values.

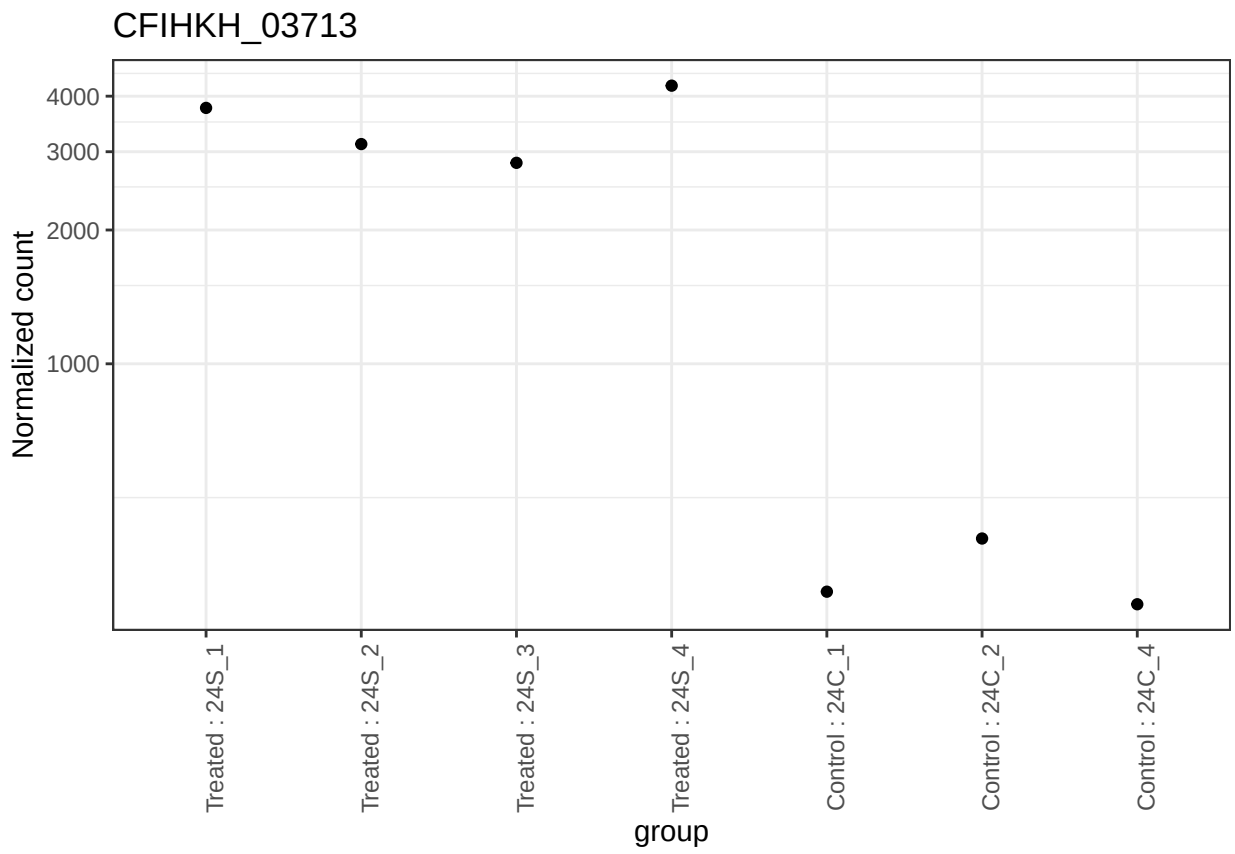
Top features

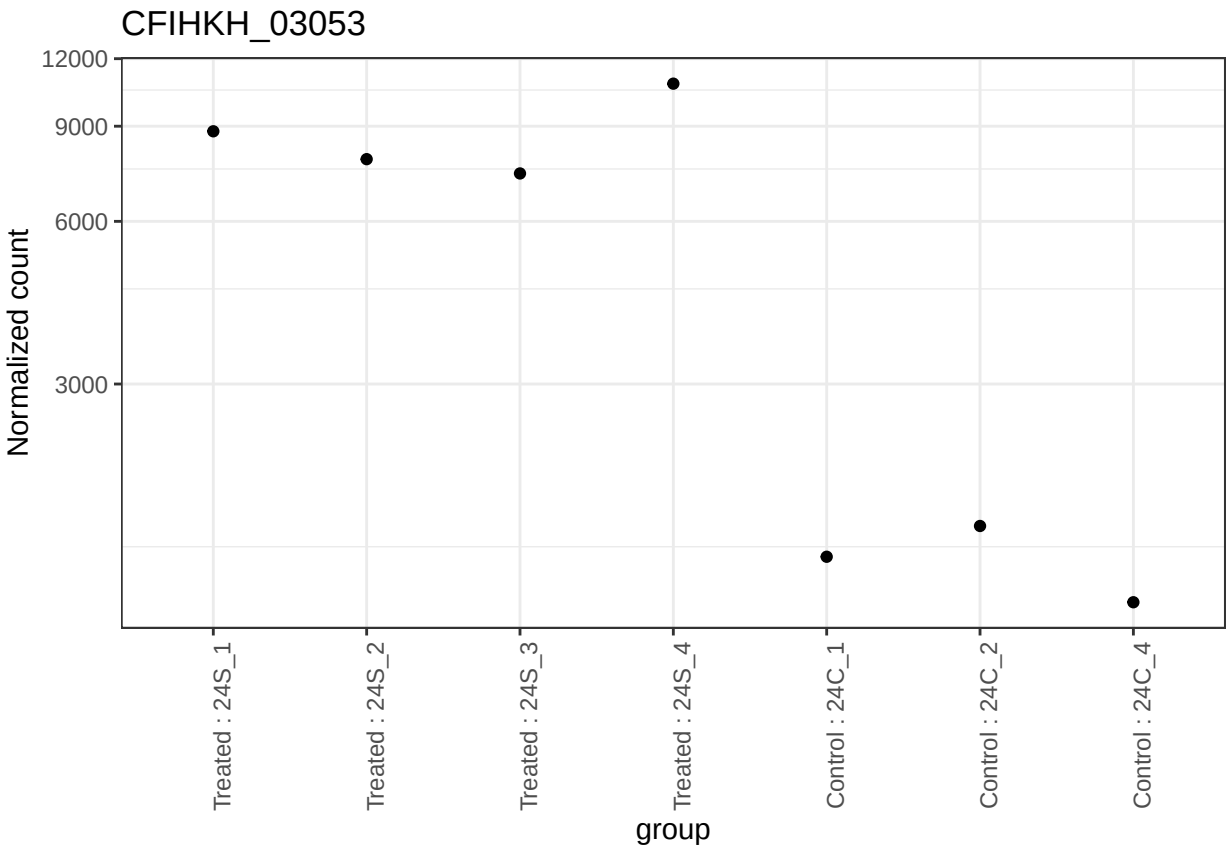
This table shows the top 500 features ordered by their BH adjusted p-values. Since the report is in PDF format, only the top 20 features are shown.

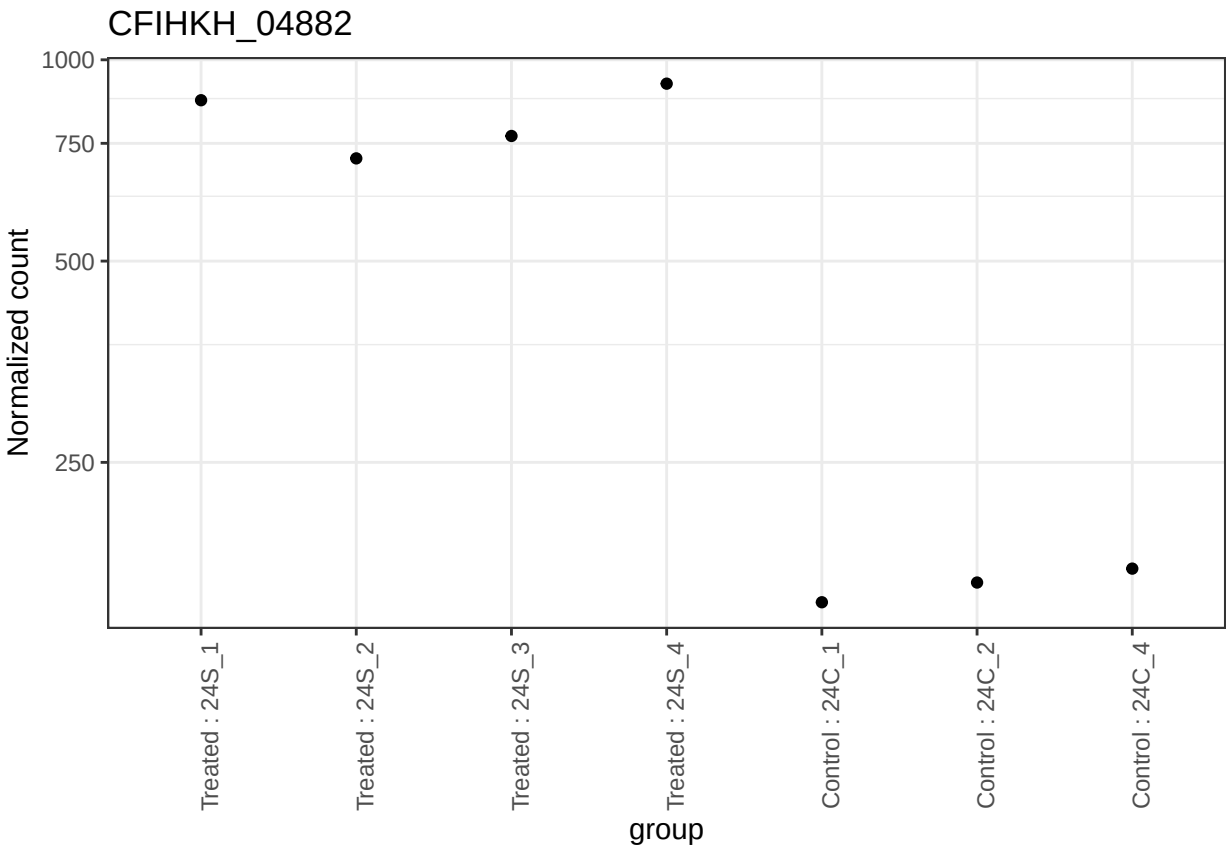
Feature	baseMean	log2FoldChange	lfcSE	pvalue	padj
CFIHKH_03713	2133.88	-3.38	0.16	3.267154e-102	1.690425e-98
CFIHKH_03053	5575.88	-2.60	0.15	1.583206e-69	4.095754e-66
CFIHKH_04882	537.44	-2.30	0.14	5.003648e-61	8.629625e-58
CFIHKH_00438	409.86	-2.01	0.13	4.893173e-52	6.329319e-49
CFIHKH_04158	354.99	-2.41	0.17	3.593476e-46	3.718529e-43
CFIHKH_03947	629.93	-2.02	0.15	9.682271e-43	8.349345e-40
CFIHKH_04421	680.86	-1.80	0.14	4.191722e-41	3.098281e-38
CFIHKH_02677	1063.80	-1.73	0.14	2.991033e-36	1.934450e-33
CFIHKH_02476	639.40	-1.51	0.13	1.913835e-33	1.100242e-30
CFIHKH_03652	745.55	-1.85	0.16	1.605958e-32	8.309224e-30
CFIHKH_03971	3304.88	-1.53	0.14	3.633282e-31	1.708964e-28
CFIHKH_01942	128.79	-1.93	0.19	1.555474e-25	6.706686e-23
CFIHKH_00221	686.69	-1.31	0.13	4.808516e-24	1.913789e-21
CFIHKH_03401	1215.24	-1.24	0.13	1.379678e-23	5.098897e-21
CFIHKH_02081	7106.32	-1.43	0.15	1.508882e-22	5.204638e-20
CFIHKH_01541	181.36	-1.60	0.17	3.674207e-22	1.188147e-19
CFIHKH_03830	463.42	-1.15	0.12	1.136336e-21	3.458471e-19
CFIHKH_03402	453.79	-1.22	0.13	1.271173e-21	3.653916e-19
CFIHKH_04632	2862.40	-1.39	0.17	3.637010e-18	9.904153e-16
CFIHKH_01476	571.85	1.07	0.14	2.482730e-16	6.422823e-14

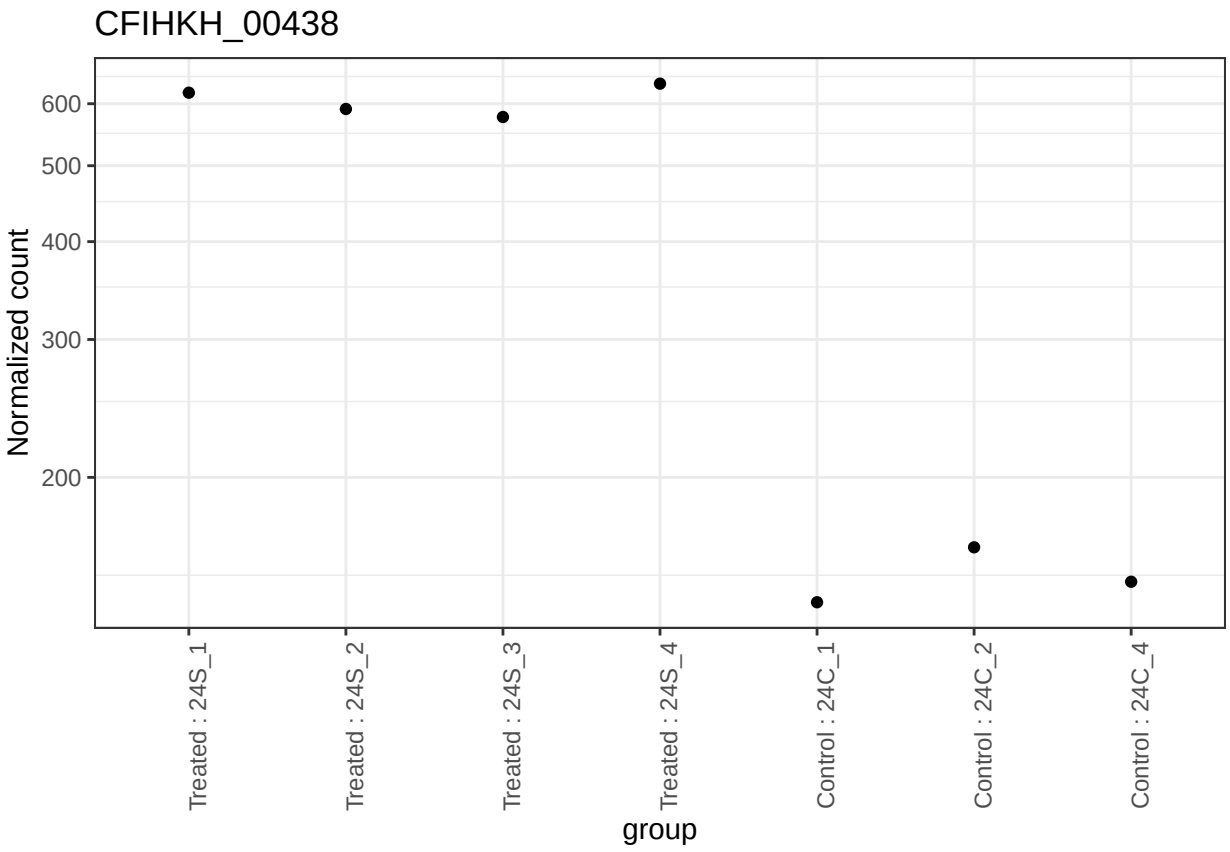
Count plots top features

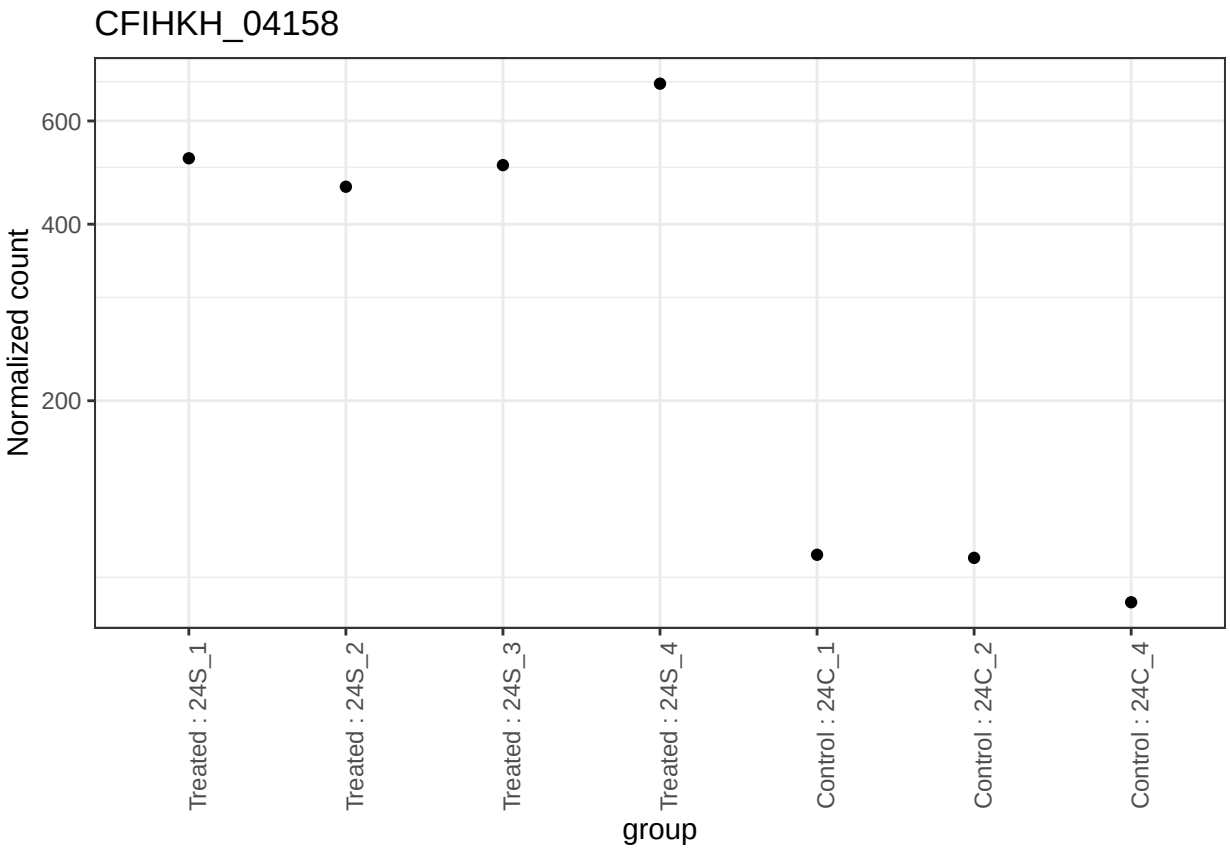
This section contains plots showing the normalized counts per sample for each group of interest. Only the best 20 features are shown, ranked by their BH adjusted p-values. The Y axis is on the log10 scale and the feature name is shown in the title of each plot.

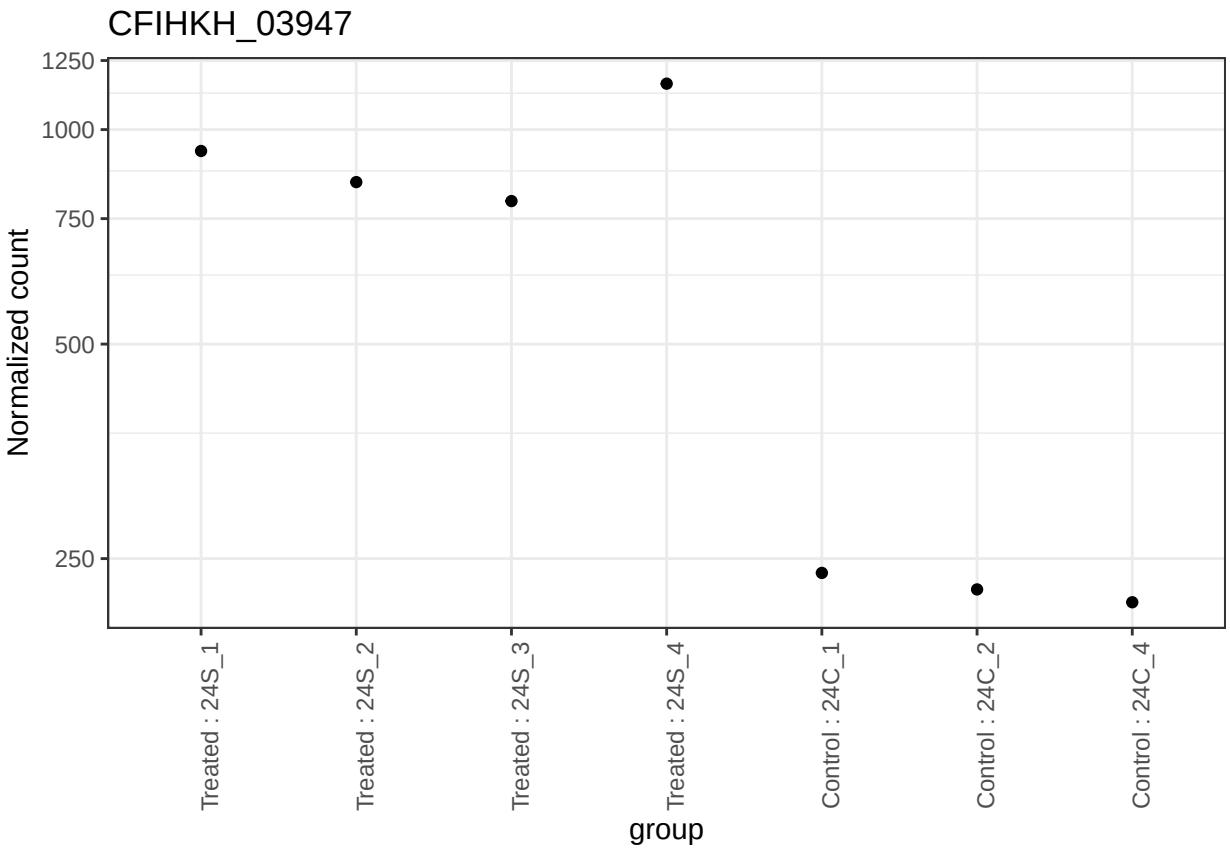

CFIHKH_03713
Normalized count
4000
3000
2000
1000
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

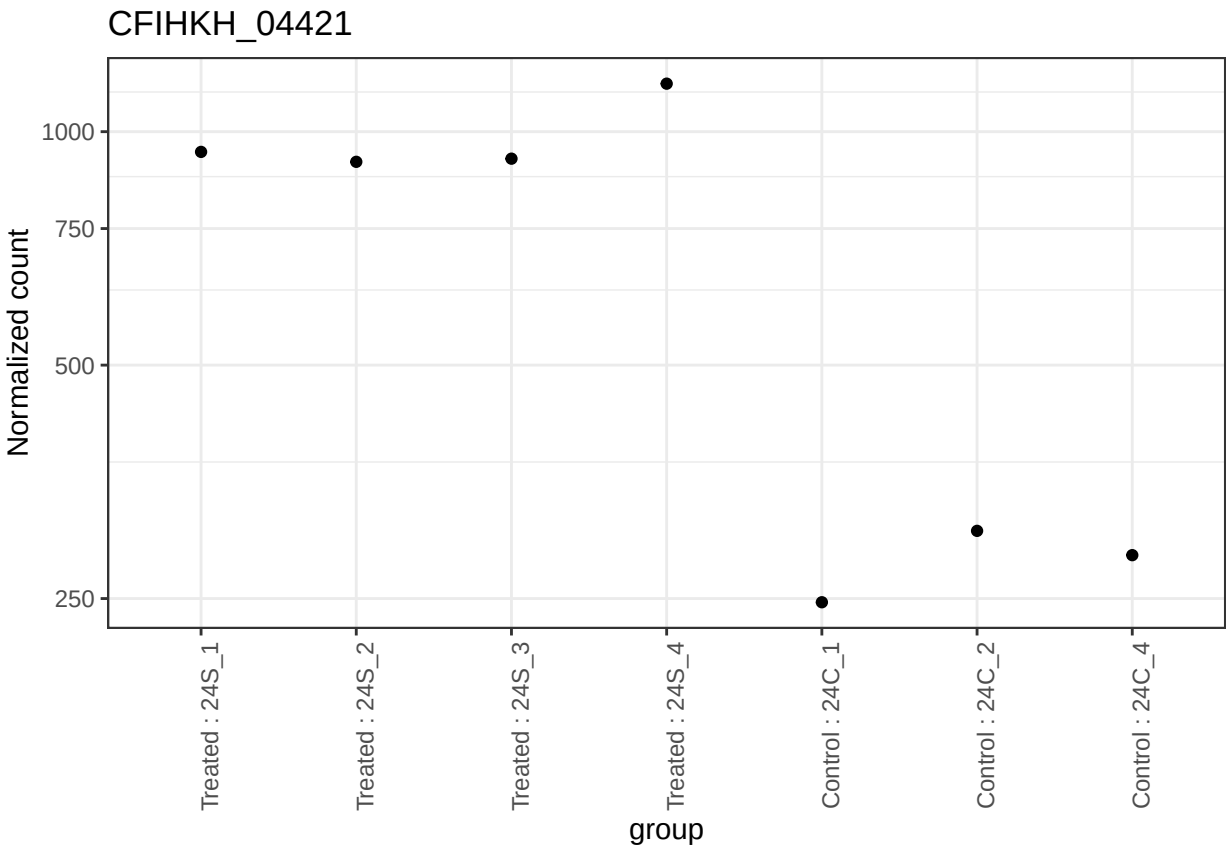

CFIHKH_03053
12000
9000
6000
3000
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

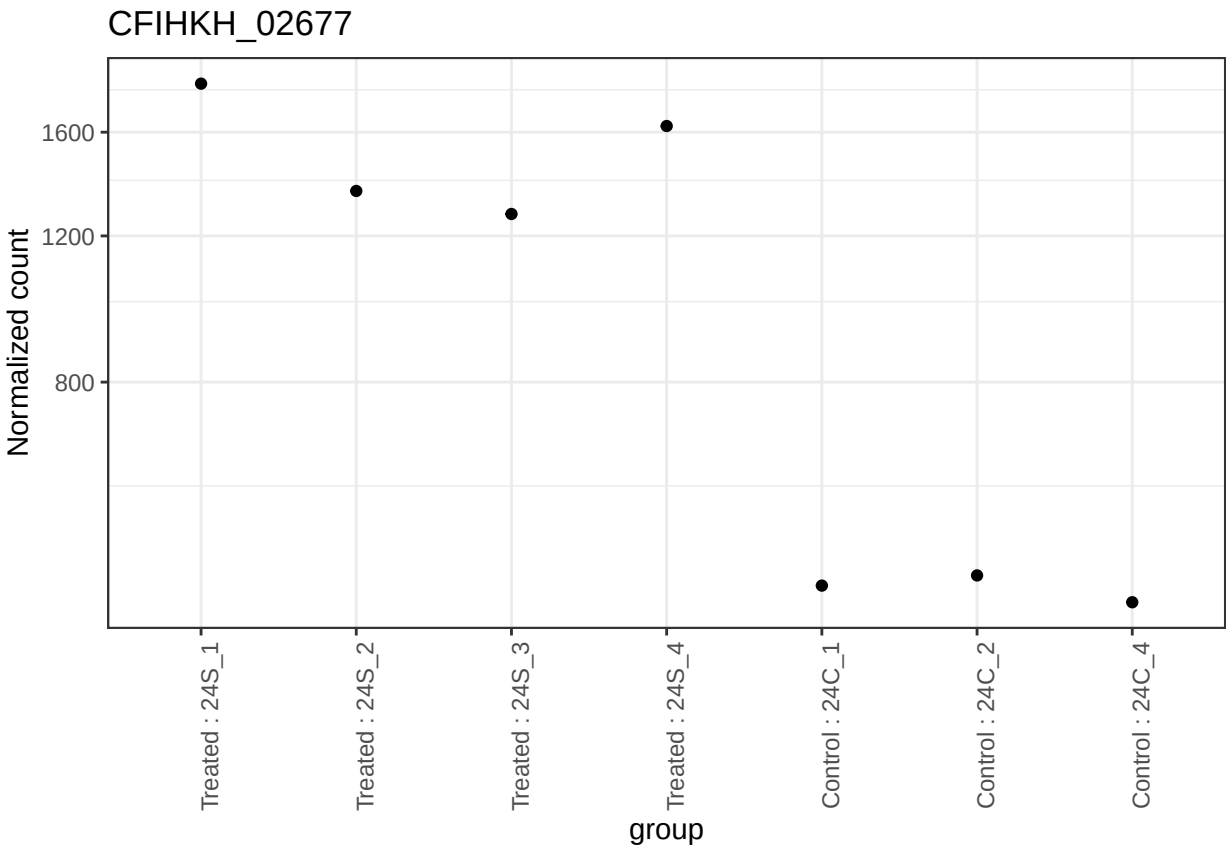

CFIHKH_04882
1000
750
500
250
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

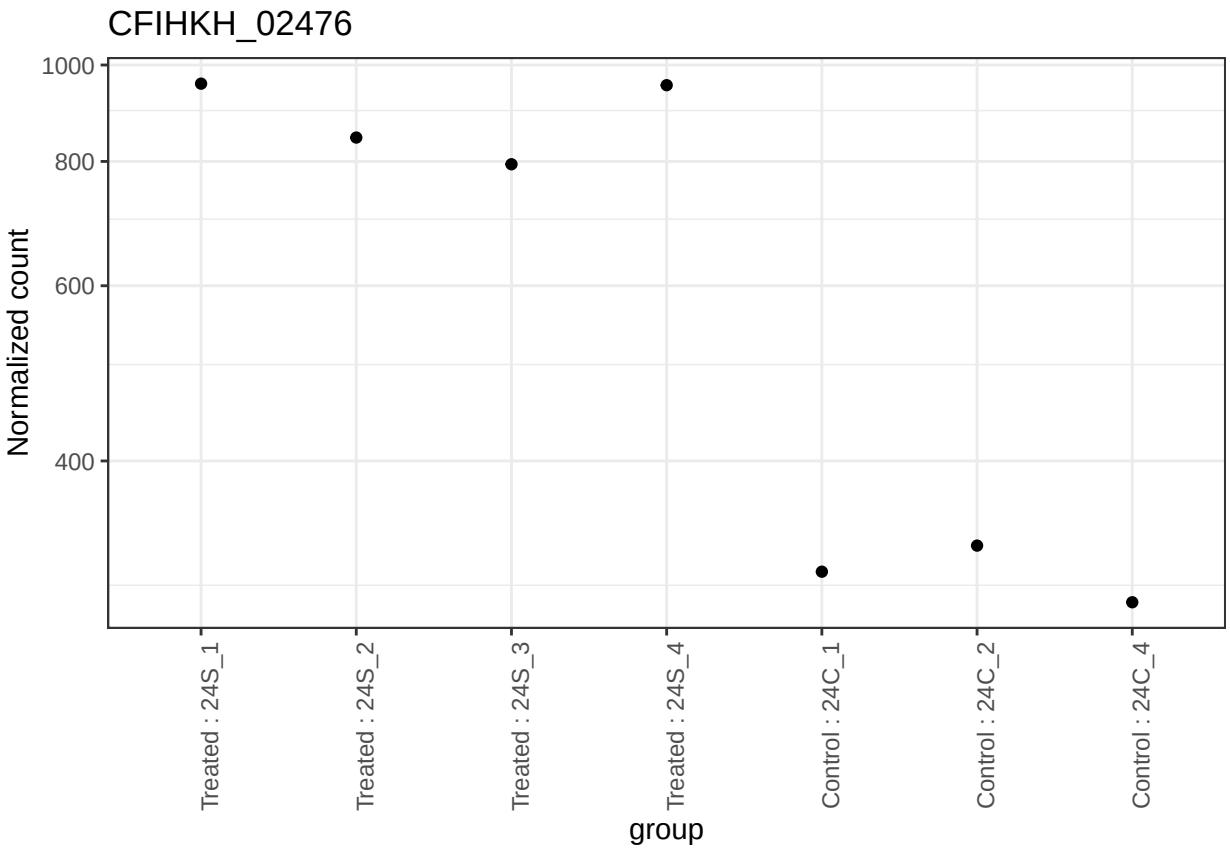

CFIHKH_00438
Normalized count
600
500
400
300
200
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

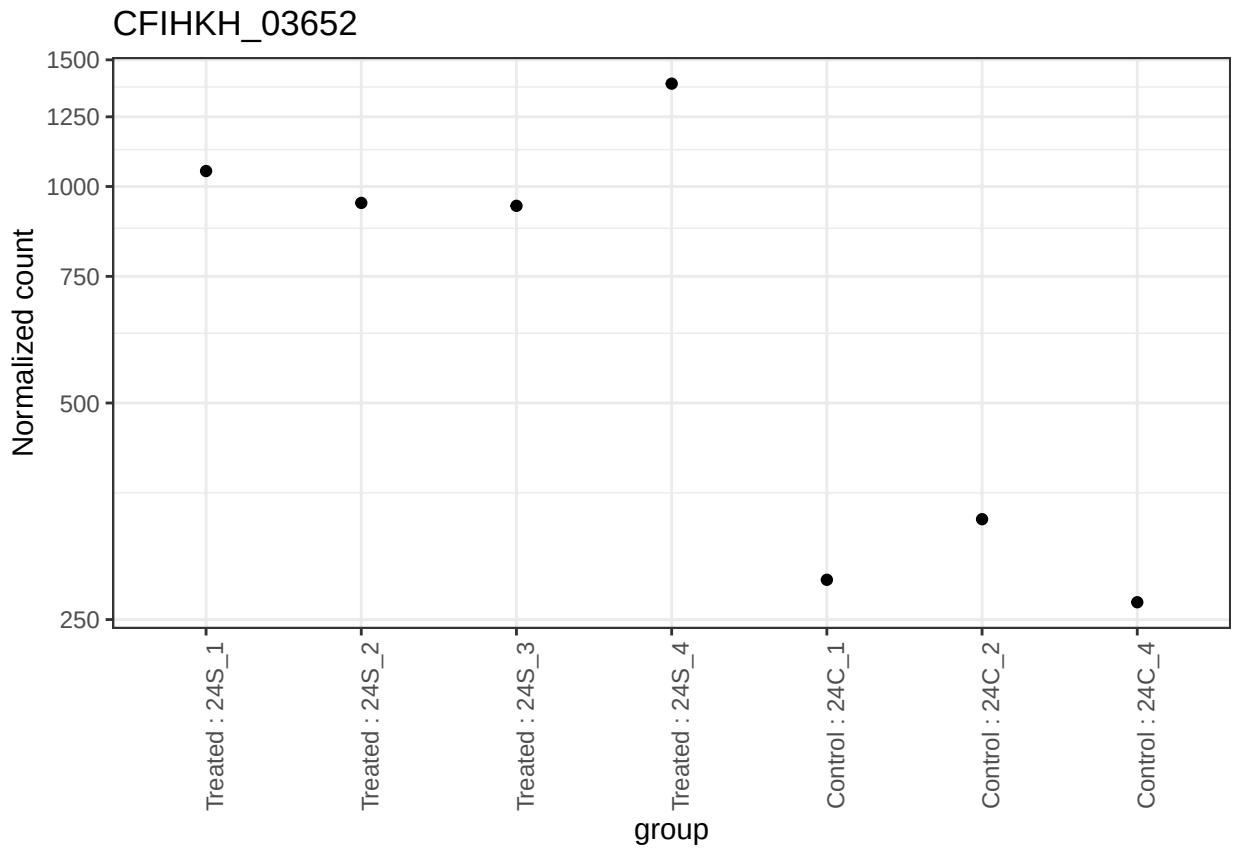

CFIHKH_04158
600
400
200
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

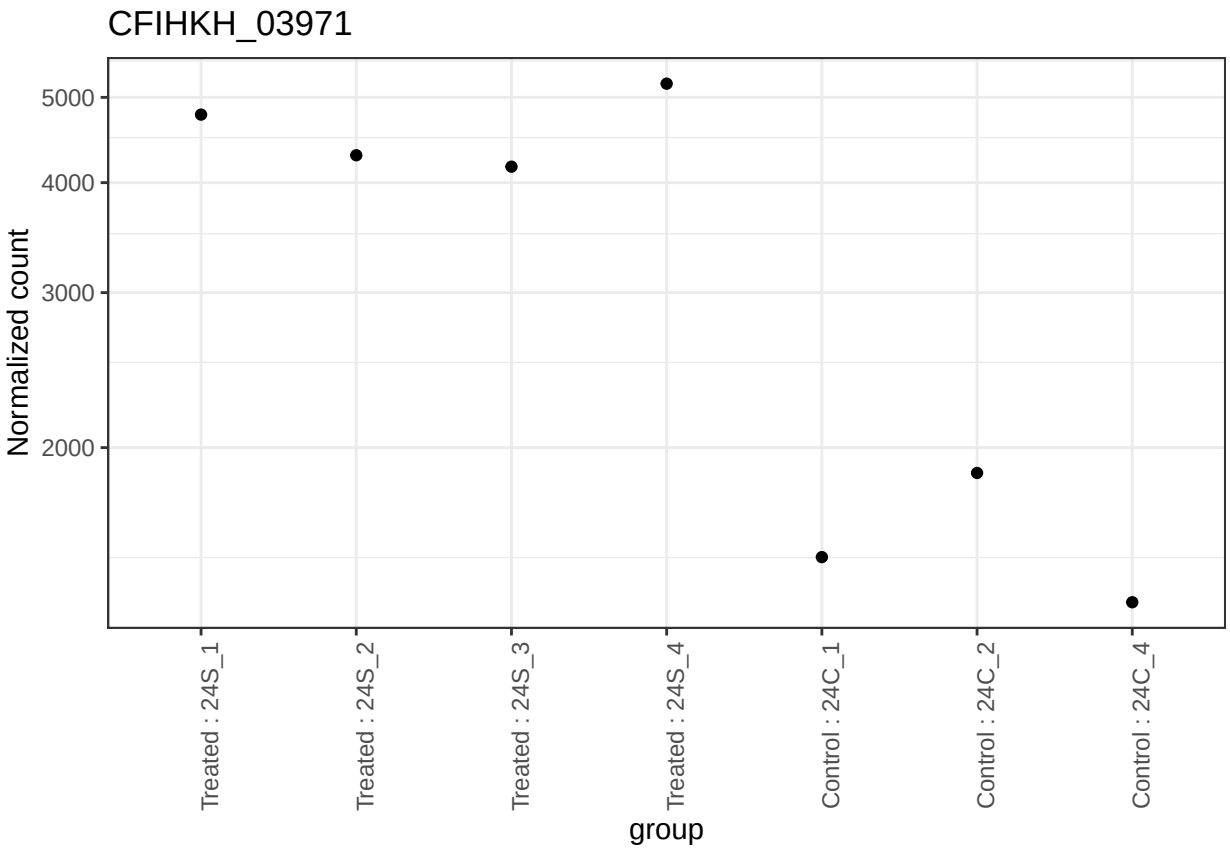

CFIHKH_03947
1250
1000
750
500
250
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

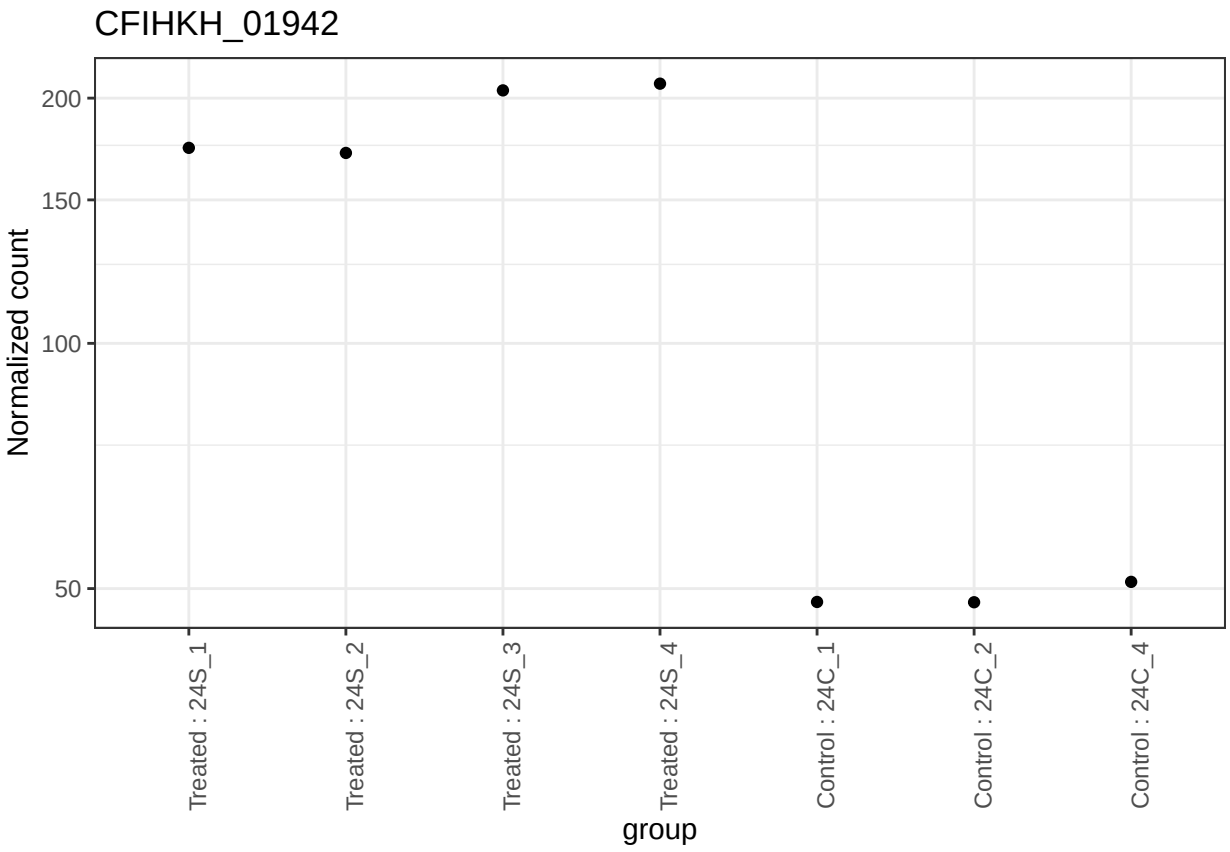

CFIHKH_04421
Normalized count
1000
750
500
250
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

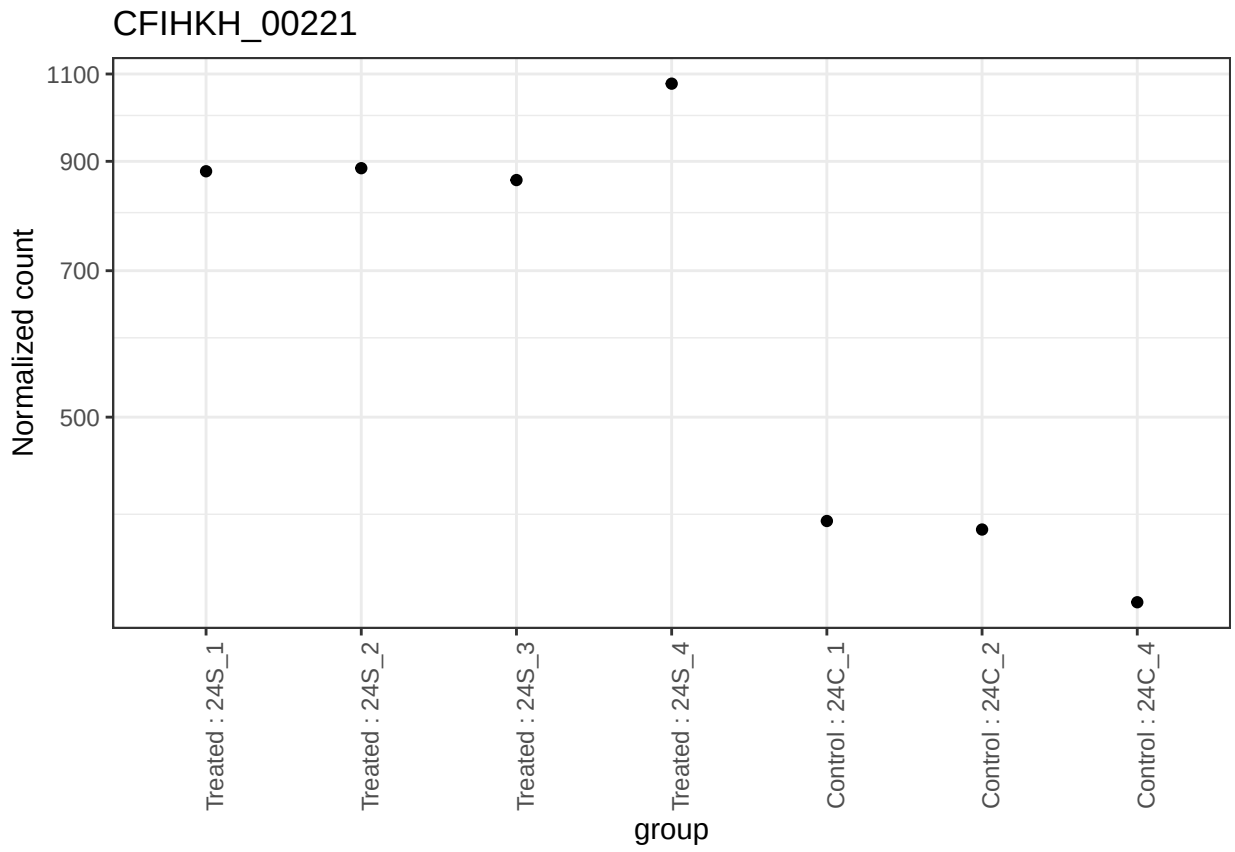

CFIHKH_02677
Normalized count
1600
1200
800
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

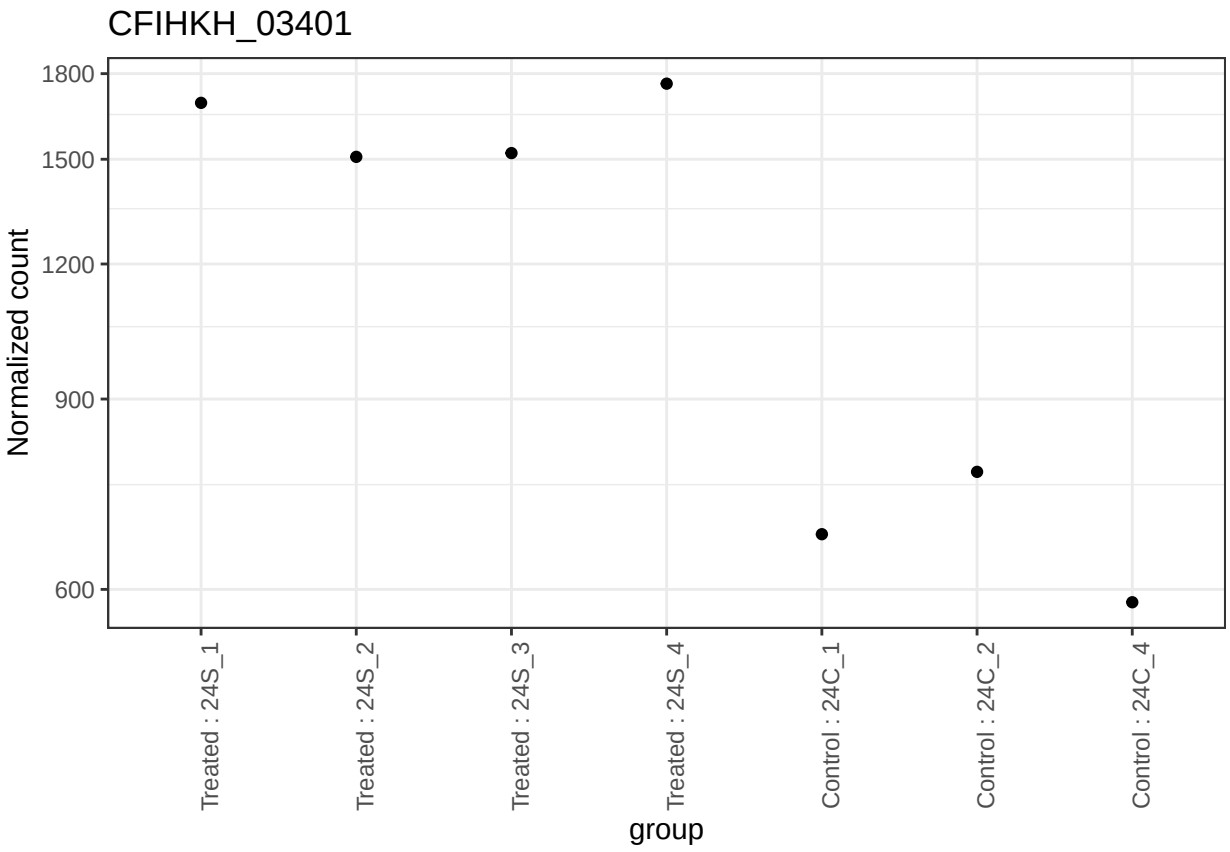

CFIHKH_02476
1000
800
600
400
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

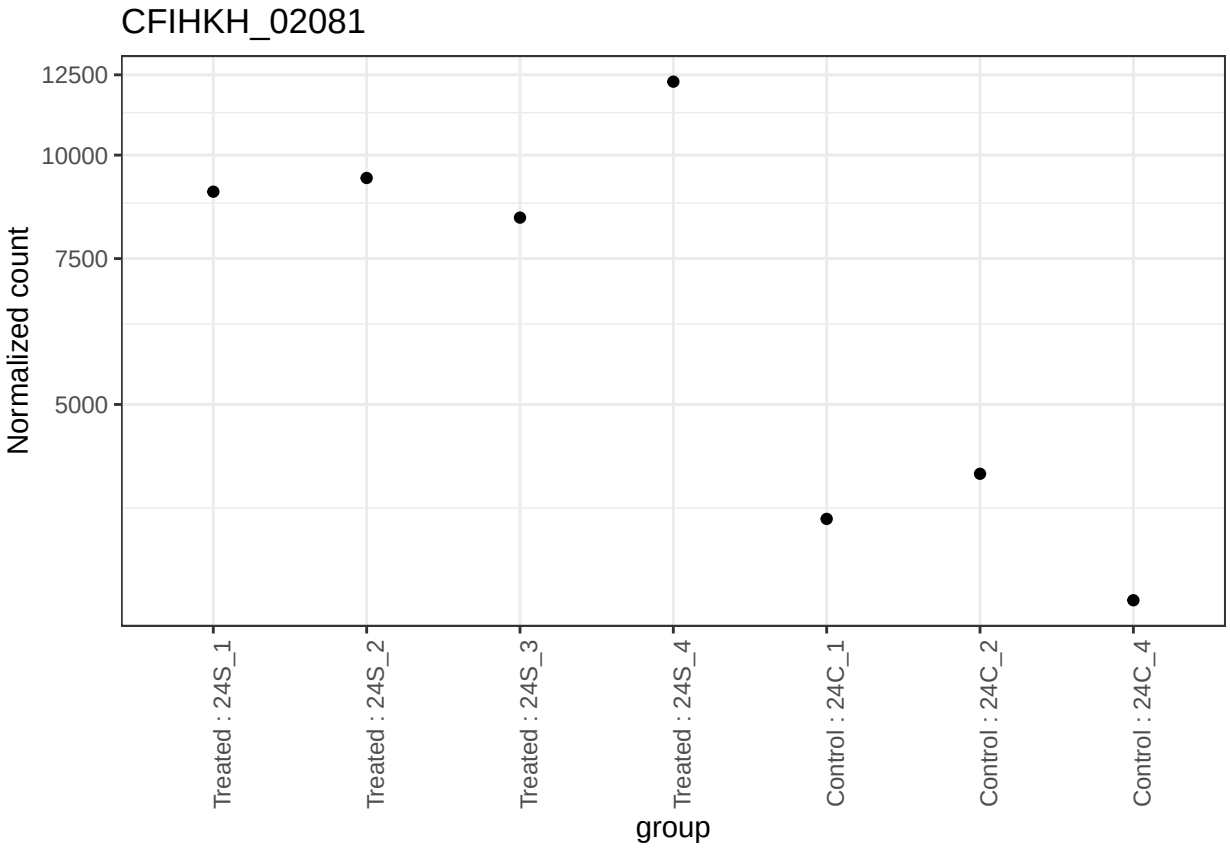

CFIHKH_03652
Normalized count
1500
1250
1000
750
500
250
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

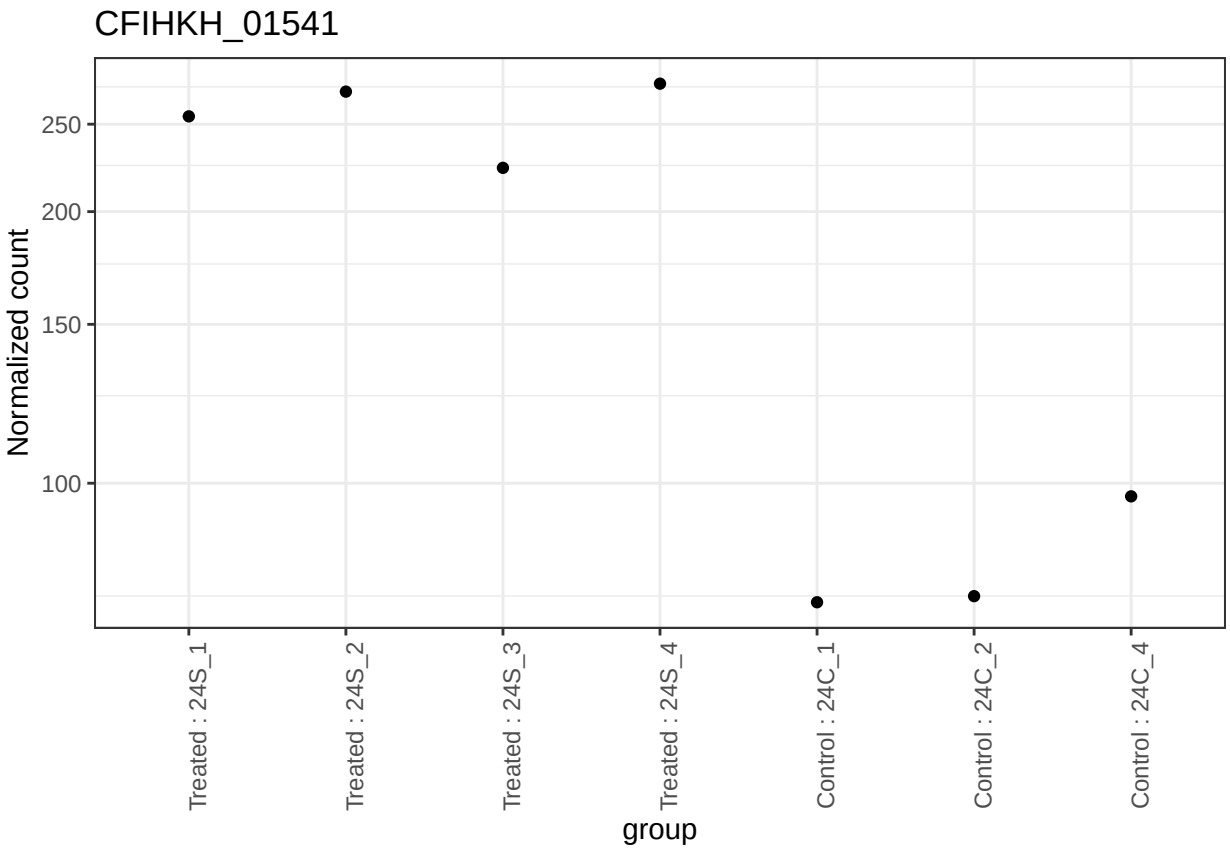

CFIHKH_03971
Normalized count
5000
4000
3000
2000
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

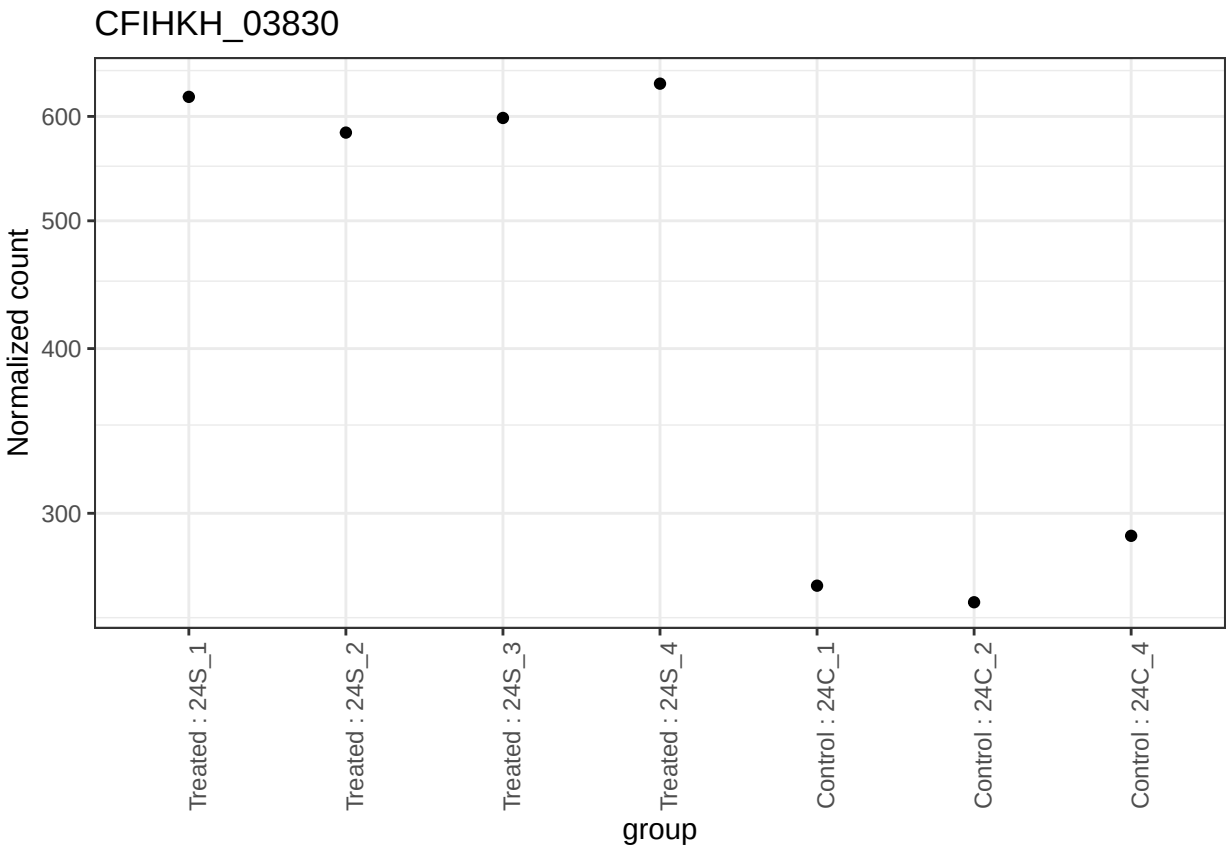

CFIHKH_01942
Normalized count
200
150
100
50
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

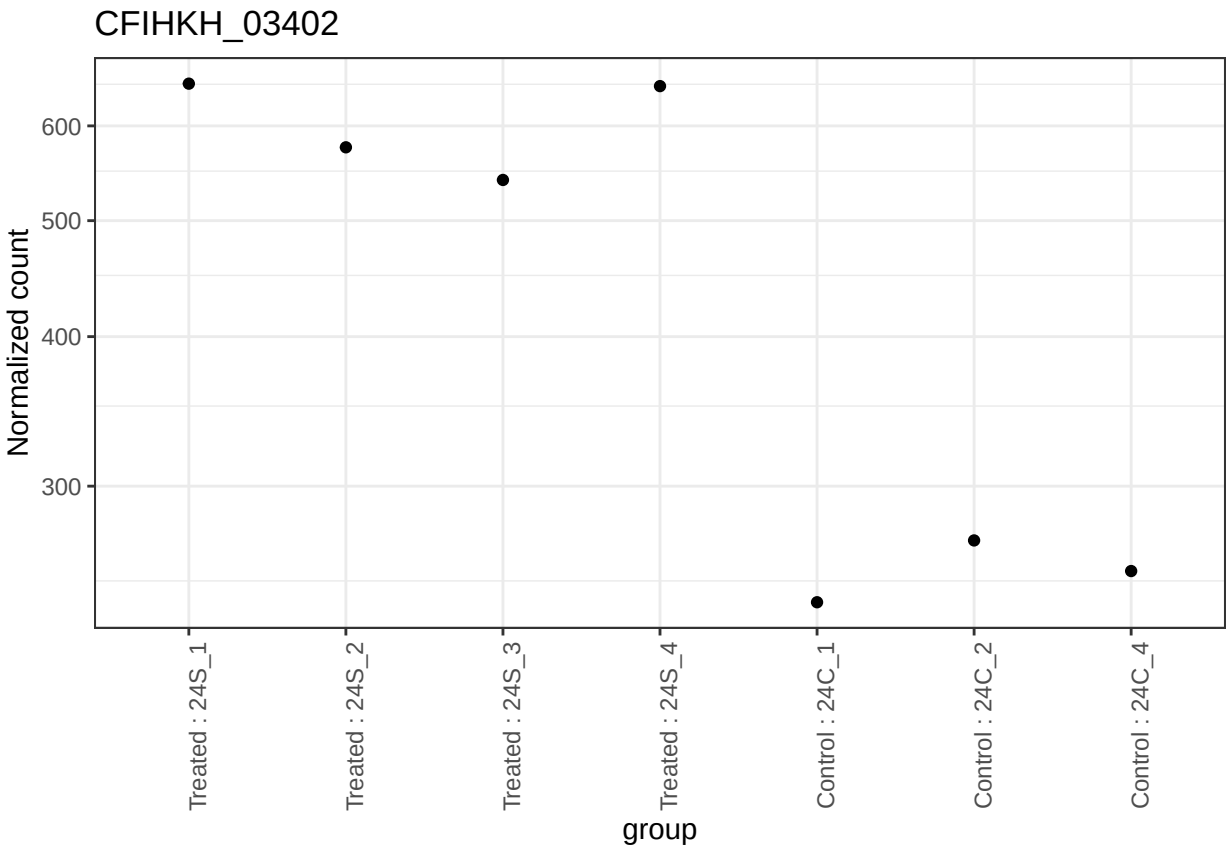

CFIHKH_00221
1100
900
700
500
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

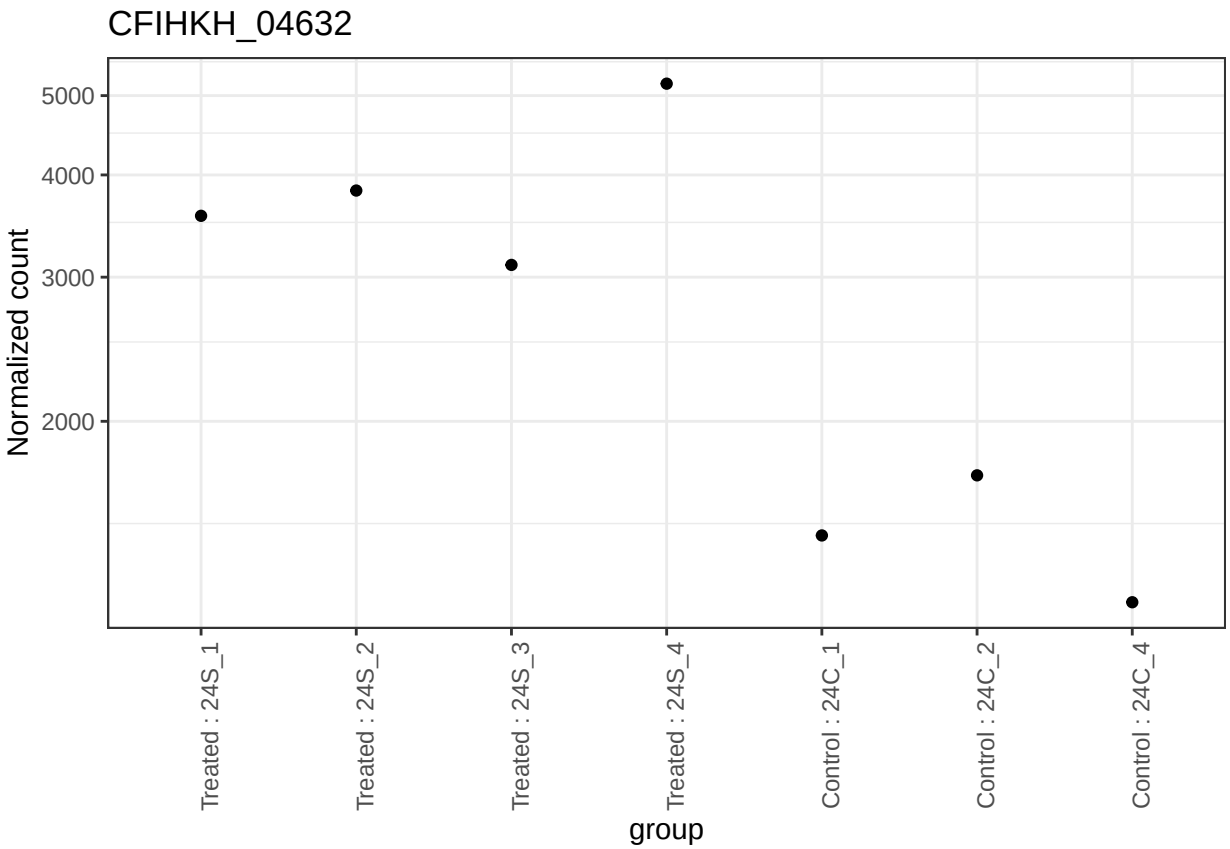

CFIHKH_03401
Normalized count
1800
1500
1200
900
600
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

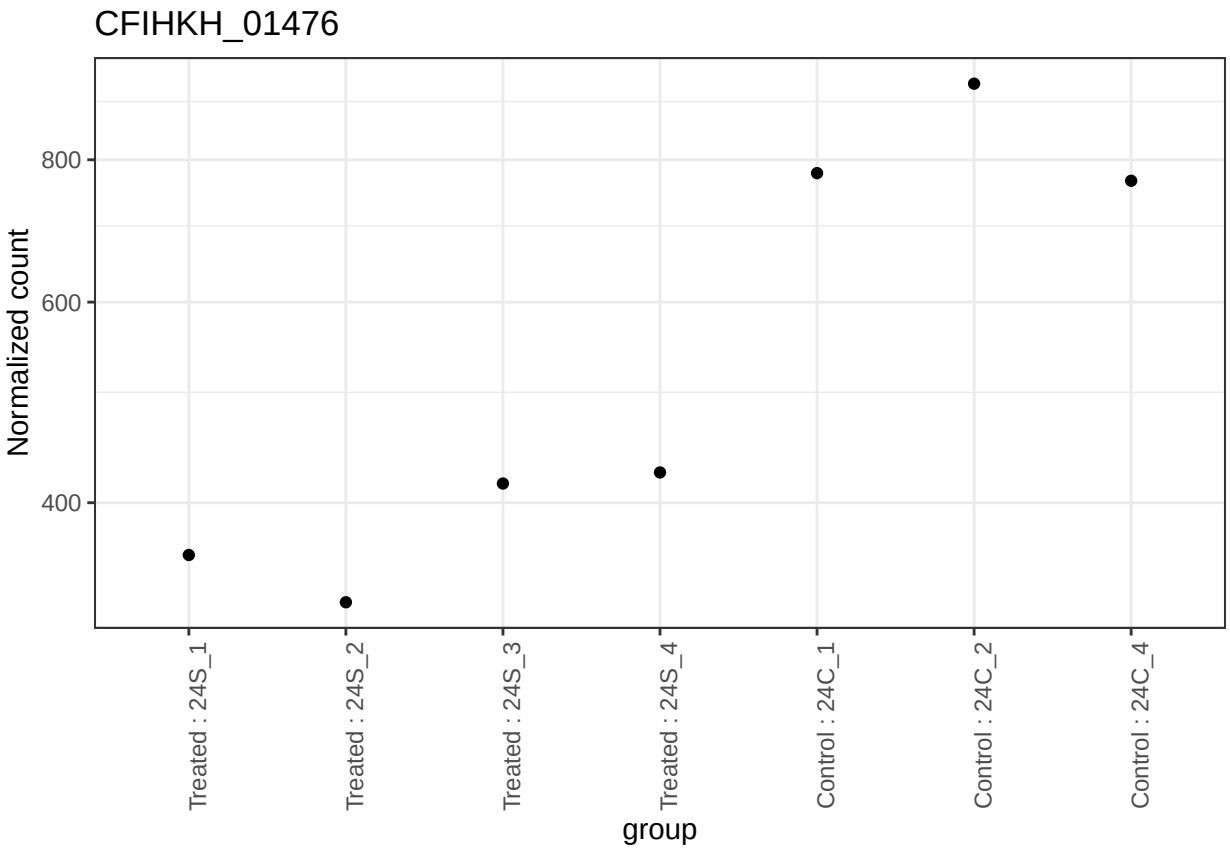

CFIHKH_02081
12500
10000
7500
5000
Normalized count
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group


CFIHKH_01541
Normalized count
250
200
150
100
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

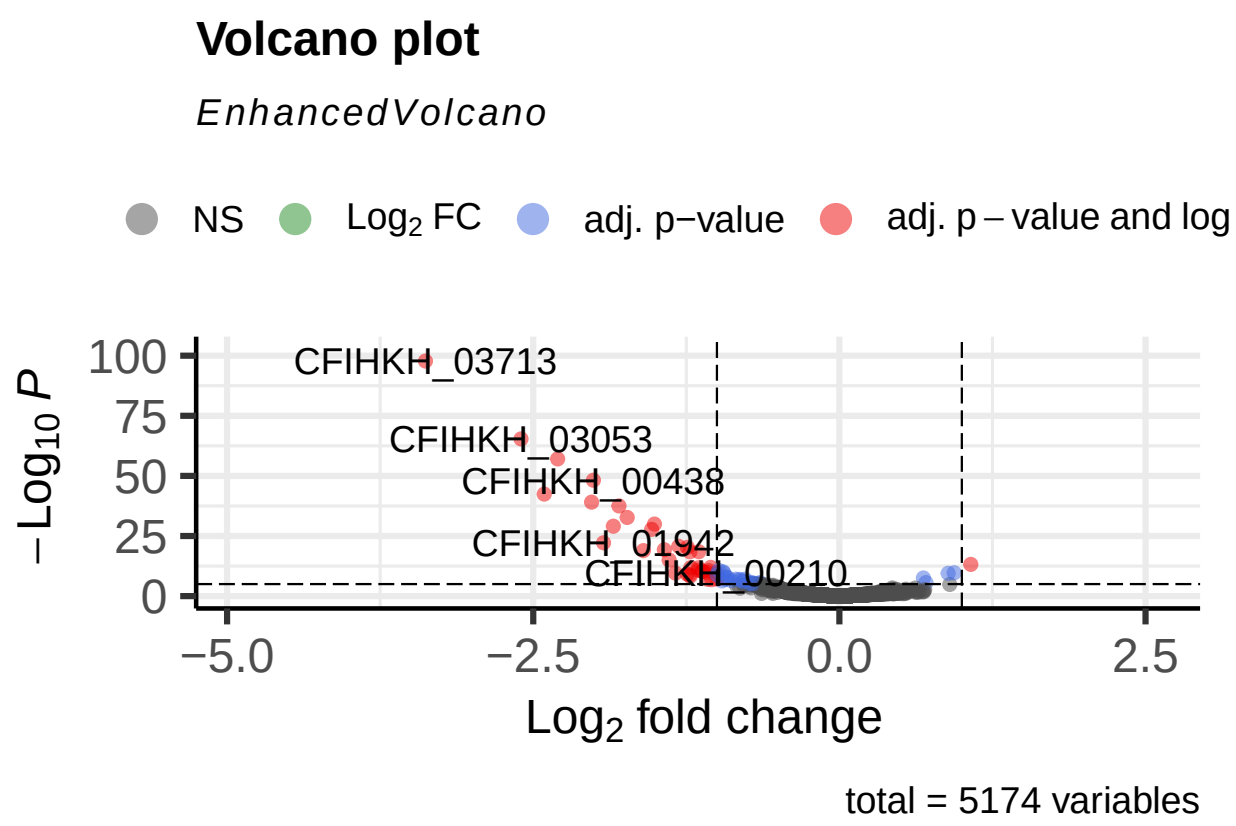

CFIHKH_03830
Normalized count
600
500
400
300
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group


CFIHKH_03402
Normalized count
600
500
400
300
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group


CFIHKH_04632
Normalized count
5000
4000
3000
2000
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group


CFIHKH_01476
Normalized count
800
600
400
Treated : 24S_1
Treated : 24S_2
Treated : 24S_3
Treated : 24S_4
Control : 24C_1
Control : 24C_2
Control : 24C_4
group

Volcano plot



The above plot shows a volcano plot (see Wikipedia) that compares the log2 fold change against the adjusted P value. The plot was created with EnhancedVolcano.

Reproducibility

The input for this report was generated with DESeq2 (Love, Huber, and Anders, 2014) using version 1.34.0 and the resulting features were called significantly differentially expressed if their BH adjusted p-values were less than `alpha` = 0.1. This report was generated in path /home/mnt/disk2/Projects/Sara_RNA_Seq/output/3_seq_box_1_24/05-CountingFilter/TPM using the following call to DESeq2Report():

```
## DESeq2Report(dds = dds, project = report.project.name, intgroup = c("condition",
##     "type"), res = deseq2.res, outdir = report.dir, output = report.output,
##     device = "pdf", template = regionReport_config, theme = theme_bw(),
##     output_format = "pdf_document")
```

Date the report was generated:

```
## [1] "2025-08-04 11:41:04 UTC"
```

Wallclock time spent generating the report:

```
## Time difference of 6.218 secs
```

`R` session information.

```
## - Session info ---------------------------------------------------------------------------------
```

```
##  setting  value
##  version  R version 4.1.3 (2022-03-10)
##  os       Debian GNU/Linux 11 (bullseye)
##  system   x86_64, linux-gnu
##  ui       X11
##  language (EN)
##  collate  C.UTF-8
##  ctype    C.UTF-8
##  tz       Etc/UTC
##  date     2025-08-04
##  pandoc   2.11 @ /opt/conda/bin/ (via rmarkdown)
##
## - Packages -----------------------------------------------------------------------------------
##  package              * version    date (UTC) lib source
##  annotate               1.72.0     2021-10-26 [1] Bioconductor
##  AnnotationDbi          1.56.1     2021-10-29 [1] Bioconductor
##  AnnotationFilter       1.18.0     2021-10-26 [1] Bioconductor
##  AnnotationForge        1.36.0     2021-10-26 [1] Bioconductor
##  apeglm                 1.16.0     2021-10-26 [1] Bioconductor
##  ash                    1.0-15     2015-09-01 [1] CRAN (R 4.1.0)
##  assertthat             0.2.1      2019-03-21 [1] CRAN (R 4.1.0)
##  backports              1.4.1      2021-12-13 [1] CRAN (R 4.1.1)
##  base64enc              0.1-3      2015-07-28 [1] CRAN (R 4.1.0)
##  bbmle                  1.0.24     2021-08-05 [1] CRAN (R 4.1.0)
##  bdsmatrix              1.3-4      2020-01-13 [1] CRAN (R 4.1.0)
##  beeswarm               0.4.0      2021-06-01 [1] CRAN (R 4.1.0)
##  Biobase              * 2.54.0     2021-10-26 [1] Bioconductor
##  BiocFileCache          2.2.0      2021-10-26 [1] Bioconductor
##  BiocGenerics         * 0.40.0     2021-10-26 [1] Bioconductor
##  BiocIO                 1.4.0      2021-10-26 [1] Bioconductor
##  BiocManager            1.30.16    2021-06-15 [1] CRAN (R 4.1.0)
##  BiocParallel           1.28.3     2021-12-09 [1] Bioconductor
##  BiocStyle            * 2.22.0     2021-10-26 [1] Bioconductor
##  biomaRt                2.50.0     2021-10-26 [1] Bioconductor
##  Biostrings             2.62.0     2021-10-26 [1] Bioconductor
##  biovizBase             1.42.0     2021-10-26 [1] Bioconductor
##  bit                    4.0.4      2020-08-04 [1] CRAN (R 4.1.0)
##  bit64                  4.0.5      2020-08-30 [1] CRAN (R 4.1.0)
##  bitops                 1.0-7      2021-04-24 [1] CRAN (R 4.1.0)
##  blob                   1.2.2      2021-07-23 [1] CRAN (R 4.1.0)
##  bookdown               0.25       2022-03-16 [1] CRAN (R 4.1.2)
##  BSgenome               1.62.0     2021-10-26 [1] Bioconductor
##  bslib                  0.3.1      2021-10-06 [1] CRAN (R 4.1.1)
##  bumphunter             1.36.0     2021-10-27 [1] Bioconductor
##  cachem                 1.0.6      2021-08-19 [1] CRAN (R 4.1.1)
##  Category               2.60.0     2021-10-26 [1] Bioconductor
##  checkmate              2.0.0      2020-02-06 [1] CRAN (R 4.1.0)
##  cli                    3.2.0      2022-02-14 [1] CRAN (R 4.1.2)
##  cluster                2.1.3      2022-03-28 [1] CRAN (R 4.1.2)
##  coda                   0.19-4     2020-09-30 [1] CRAN (R 4.1.0)
##  codetools              0.2-18     2020-11-04 [1] CRAN (R 4.1.0)
##  colorspace             2.0-3      2022-02-21 [1] CRAN (R 4.1.2)
##  crayon                 1.5.1      2022-03-26 [1] CRAN (R 4.1.2)
##  crosstalk              1.2.0      2021-11-04 [1] CRAN (R 4.1.1)
```

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##  curl                  4.3.2      2021-06-23 [1] CRAN (R 4.1.0)
##  data.table            1.14.2     2021-09-27 [1] CRAN (R 4.1.1)
##  DBI                   1.1.2      2021-12-20 [1] CRAN (R 4.1.2)
##  dbplyr                2.1.1      2021-04-06 [1] CRAN (R 4.1.0)
##  DEFormats             1.22.0     2021-10-26 [1] Bioconductor
##  DelayedArray          0.20.0     2021-10-26 [1] Bioconductor
##  derfinder             1.28.0     2021-10-26 [1] Bioconductor
##  derfinderHelper       1.28.0     2021-10-26 [1] Bioconductor
##  DESeq2              * 1.34.0     2021-10-26 [1] Bioconductor
##  dichromat             2.0-0      2013-01-24 [1] CRAN (R 4.1.0)
##  digest                0.6.29     2021-12-01 [1] CRAN (R 4.1.1)
##  doParallel            1.0.17     2022-02-07 [1] CRAN (R 4.1.2)
##  doRNG                 1.8.2      2020-01-27 [1] CRAN (R 4.1.0)
##  dplyr                 1.0.8      2022-02-08 [1] CRAN (R 4.1.2)
##  DT                  * 0.22       2022-03-28 [1] CRAN (R 4.1.2)
##  edgeR                 3.36.0     2021-10-26 [1] Bioconductor
##  ellipsis              0.3.2      2021-04-29 [1] CRAN (R 4.1.0)
##  emdbook               1.3.12     2020-02-19 [1] CRAN (R 4.1.0)
##  EnhancedVolcano     * 1.12.0     2021-10-26 [1] Bioconductor
##  ensembldb             2.18.1     2021-11-03 [1] Bioconductor
##  evaluate              0.15       2022-02-18 [1] CRAN (R 4.1.2)
##  extrafont             0.17       2014-12-08 [1] CRAN (R 4.1.0)
##  extrafontdb           1.0        2012-06-11 [1] CRAN (R 4.1.0)
##  fansi                 1.0.3      2022-03-24 [1] CRAN (R 4.1.2)
##  farver                2.1.0      2021-02-28 [1] CRAN (R 4.1.0)
##  fastmap               1.1.0      2021-01-25 [1] CRAN (R 4.1.0)
##  filelock              1.0.2      2018-10-05 [1] CRAN (R 4.1.0)
##  foreach               1.5.2      2022-02-02 [1] CRAN (R 4.1.2)
##  foreign               0.8-82     2022-01-13 [1] CRAN (R 4.1.2)
##  Formula               1.2-4      2020-10-16 [1] CRAN (R 4.1.0)
##  genefilter            1.76.0     2021-10-26 [1] Bioconductor
##  geneplotter           1.72.0     2021-10-26 [1] Bioconductor
##  generics              0.1.2      2022-01-31 [1] CRAN (R 4.1.2)
##  GenomeInfoDb        * 1.30.0     2021-10-26 [1] Bioconductor
##  GenomeInfoDbData      1.2.7      2022-03-31 [1] Bioconductor
##  GenomicAlignments     1.30.0     2021-10-26 [1] Bioconductor
##  GenomicFeatures       1.46.1     2021-10-27 [1] Bioconductor
##  GenomicFiles          1.30.0     2021-10-26 [1] Bioconductor
##  GenomicRanges       * 1.46.1     2021-11-18 [1] Bioconductor
##  GGally                2.1.2      2021-06-21 [1] CRAN (R 4.1.0)
##  ggalt                 0.4.0      2017-02-15 [1] CRAN (R 4.1.0)
##  ggbeeswarm            0.6.0      2017-08-07 [1] CRAN (R 4.1.0)
##  ggbio                 1.42.0     2021-11-02 [1] Bioconductor
##  ggplot2             * 3.3.5      2021-06-25 [1] CRAN (R 4.1.0)
##  ggrastr               1.0.1      2021-12-08 [1] CRAN (R 4.1.1)
##  ggrepel             * 0.9.1      2021-01-15 [1] CRAN (R 4.1.0)
##  glue                  1.6.2      2022-02-24 [1] CRAN (R 4.1.2)
##  GO.db                 3.14.0     2022-03-31 [1] Bioconductor
##  GOstats               2.60.0     2021-10-26 [1] Bioconductor
##  graph                 1.72.0     2021-10-26 [1] Bioconductor
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##  GSEABase              1.56.0     2021-10-26 [1] Bioconductor
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##  highr                 0.9        2021-04-16 [1] CRAN (R 4.1.0)
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##  hms                   1.1.1      2021-09-26 [1] CRAN (R 4.1.1)
##  htmlTable             2.4.0      2022-01-04 [1] CRAN (R 4.1.2)
##  htmltools             0.5.2      2021-08-25 [1] CRAN (R 4.1.1)
##  htmlwidgets           1.5.4      2021-09-08 [1] CRAN (R 4.1.1)
##  httr                  1.4.2      2020-07-20 [1] CRAN (R 4.1.0)
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##  IRanges             * 2.28.0     2021-10-26 [1] Bioconductor
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##  jpeg                  0.1-9      2021-07-24 [1] CRAN (R 4.1.1)
##  jquerylib             0.1.4      2021-04-26 [1] CRAN (R 4.1.0)
##  jsonlite              1.8.0      2022-02-22 [1] CRAN (R 4.1.2)
##  KEGGREST              1.34.0     2021-10-26 [1] Bioconductor
##  KernSmooth            2.23-20    2021-05-03 [1] CRAN (R 4.1.0)
##  knitcitations       * 1.0.12     2021-01-10 [1] CRAN (R 4.1.0)
##  knitr               * 1.38       2022-03-25 [1] CRAN (R 4.1.2)
##  knitrBootstrap        1.0.2      2018-05-24 [1] CRAN (R 4.1.0)
##  labeling              0.4.2      2020-10-20 [1] CRAN (R 4.1.2)
##  lattice               0.20-45    2021-09-22 [1] CRAN (R 4.1.1)
##  latticeExtra          0.6-29     2019-12-19 [1] CRAN (R 4.1.0)
##  lazyeval              0.2.2      2019-03-15 [1] CRAN (R 4.1.0)
##  lifecycle             1.0.1      2021-09-24 [1] CRAN (R 4.1.1)
##  limma                 3.50.1     2022-02-17 [1] Bioconductor
##  locfit                1.5-9.5    2022-03-03 [1] CRAN (R 4.1.2)
##  lubridate             1.8.0      2021-10-07 [1] CRAN (R 4.1.1)
##  magrittr              2.0.3      2022-03-30 [1] CRAN (R 4.1.3)
##  maps                  3.4.0      2021-09-25 [1] CRAN (R 4.1.1)
##  markdown              1.1        2019-08-07 [1] CRAN (R 4.1.0)
##  MASS                  7.3-56     2022-03-23 [1] CRAN (R 4.1.2)
##  Matrix                1.4-1      2022-03-23 [1] CRAN (R 4.1.2)
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##  matrixStats         * 0.61.0     2021-09-17 [1] CRAN (R 4.1.1)
##  memoise               2.0.1      2021-11-26 [1] CRAN (R 4.1.1)
##  munsell               0.5.0      2018-06-12 [1] CRAN (R 4.1.2)
##  mvtnorm               1.1-3      2021-10-08 [1] CRAN (R 4.1.1)
##  nnet                  7.3-17     2022-01-13 [1] CRAN (R 4.1.2)
##  numDeriv              2016.8-1.1 2019-06-06 [1] CRAN (R 4.1.0)
##  openxlsx            * 4.2.5      2021-12-14 [1] CRAN (R 4.1.1)
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##  PFAM.db               3.14.0     2022-03-31 [1] Bioconductor
##  pheatmap            * 1.0.12     2019-01-04 [1] CRAN (R 4.1.0)
##  pillar                1.7.0      2022-02-01 [1] CRAN (R 4.1.2)
##  pkgconfig             2.0.3      2019-09-22 [1] CRAN (R 4.1.0)
##  plyr                  1.8.7      2022-03-24 [1] CRAN (R 4.1.2)
##  png                   0.1-7      2013-12-03 [1] CRAN (R 4.1.0)
##  prettyunits           1.1.1      2020-01-24 [1] CRAN (R 4.1.0)
##  progress              1.2.2      2019-05-16 [1] CRAN (R 4.1.0)
##  proj4                 1.0-10.1   2021-01-26 [1] CRAN (R 4.1.1)
##  ProtGenerics          1.26.0     2021-10-26 [1] Bioconductor
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##  qvalue                2.26.0     2021-10-26 [1] Bioconductor
##  R.methodsS3           1.8.1      2020-08-26 [1] CRAN (R 4.1.0)
##  R.oo                  1.24.0     2020-08-26 [1] CRAN (R 4.1.0)
##  R.utils               2.11.0     2021-09-26 [1] CRAN (R 4.1.1)
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##  R6                    2.5.1      2021-08-19 [1] CRAN (R 4.1.1)
##  ragg                  1.2.2      2022-02-21 [1] CRAN (R 4.1.2)
##  rappdirs              0.3.3      2021-01-31 [1] CRAN (R 4.1.0)
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##  RColorBrewer        * 1.1-2      2014-12-07 [1] CRAN (R 4.1.0)
##  Rcpp                  1.0.8.3    2022-03-17 [1] CRAN (R 4.1.2)
##  RCurl                 1.98-1.6   2022-02-08 [1] CRAN (R 4.1.2)
##  RefManageR            1.3.0      2020-11-13 [1] CRAN (R 4.1.0)
##  regionReport        * 1.27.1     2021-08-05 [1] Bioconductor
##  ReportingTools      * 2.34.0     2021-10-26 [1] Bioconductor
##  reshape               0.8.8      2018-10-23 [1] CRAN (R 4.1.2)
##  reshape2              1.4.4      2020-04-09 [1] CRAN (R 4.1.0)
##  restfulr              0.0.13     2017-08-06 [1] CRAN (R 4.1.2)
##  Rgraphviz             2.38.0     2021-10-26 [1] Bioconductor
##  rjson                 0.2.21     2022-01-09 [1] CRAN (R 4.1.2)
##  rlang                 1.0.2      2022-03-04 [1] CRAN (R 4.1.2)
##  rmarkdown             2.13       2022-03-10 [1] CRAN (R 4.1.2)
##  rngtools              1.5.2      2021-09-20 [1] CRAN (R 4.1.1)
##  rpart                 4.1.16     2022-01-24 [1] CRAN (R 4.1.2)
##  Rsamtools             2.10.0     2021-10-26 [1] Bioconductor
##  RSQLite               2.2.8      2021-08-21 [1] CRAN (R 4.1.1)
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##  rtracklayer           1.54.0     2021-10-26 [1] Bioconductor
##  Rttf2pt1              1.3.10     2022-02-07 [1] CRAN (R 4.1.2)
##  S4Vectors           * 0.32.3     2021-11-21 [1] Bioconductor
##  sass                  0.4.1      2022-03-23 [1] CRAN (R 4.1.2)
##  scales                1.1.1      2020-05-11 [1] CRAN (R 4.1.0)
##  sessioninfo         * 1.2.2      2021-12-06 [1] CRAN (R 4.1.1)
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##  stringr               1.4.0      2019-02-10 [1] CRAN (R 4.1.0)
##  SummarizedExperiment * 1.24.0    2021-10-26 [1] Bioconductor
##  survival              3.3-1      2022-03-03 [1] CRAN (R 4.1.2)
##  svglite               2.1.0      2022-02-03 [1] CRAN (R 4.1.2)
##  systemfonts           1.0.4      2022-02-11 [1] CRAN (R 4.1.2)
##  textshaping           0.3.6      2021-10-13 [1] CRAN (R 4.1.3)
##  tibble                3.1.6      2021-11-07 [1] CRAN (R 4.1.1)
##  tidyselect            1.1.2      2022-02-21 [1] CRAN (R 4.1.2)
##  utf8                  1.2.2      2021-07-24 [1] CRAN (R 4.1.0)
##  VariantAnnotation     1.40.0     2021-10-26 [1] Bioconductor
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##  vipor                 0.4.5      2017-03-22 [1] CRAN (R 4.1.0)
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##  xfun                  0.30       2022-03-02 [1] CRAN (R 4.1.2)
##  XML                   3.99-0.9   2022-02-24 [1] CRAN (R 4.1.2)
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##  zlibbioc              1.40.0     2021-10-26 [1] Bioconductor
##
##  [1] /opt/conda/lib/R/library
##
## ------------------------------------------------------------------------------------------------------
```

Pandoc version used: 2.11.

Bibliography

This report was created with `regionReport` (Collado-Torres, Jaffe, and Leek, 2016) using `rmarkdown` (Allaire, Xie, McPherson, et al., 2022) while `knitr` (Xie, 2014) and `DT` (Xie, Cheng, and Tan, 2022) were running behind the scenes. `pheatmap` (Kolde, 2019) was used to create the sample distances heatmap. Several plots were made with `ggplot2` (Wickham, 2016). The volcano plot was create with `EnhancedVolcano` (Blighe, Rana, and Lewis, 2018).

Citations made with `knitcitations` . The BibTeX file can be found here. [1] J. Allaire, Y. Xie, J. McPherson, et al. *rmarkdown: Dynamic Documents for R*. R package version 2.13. 2022. <URL: https://github.com/rstudio/rmarkdown>.

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